

Team 06:

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Github Repo URL: https://github.com/adi776borate/CS203-STT-Labs/tree/main/CS203_Lab_08

Docker Hub Repo URLs:

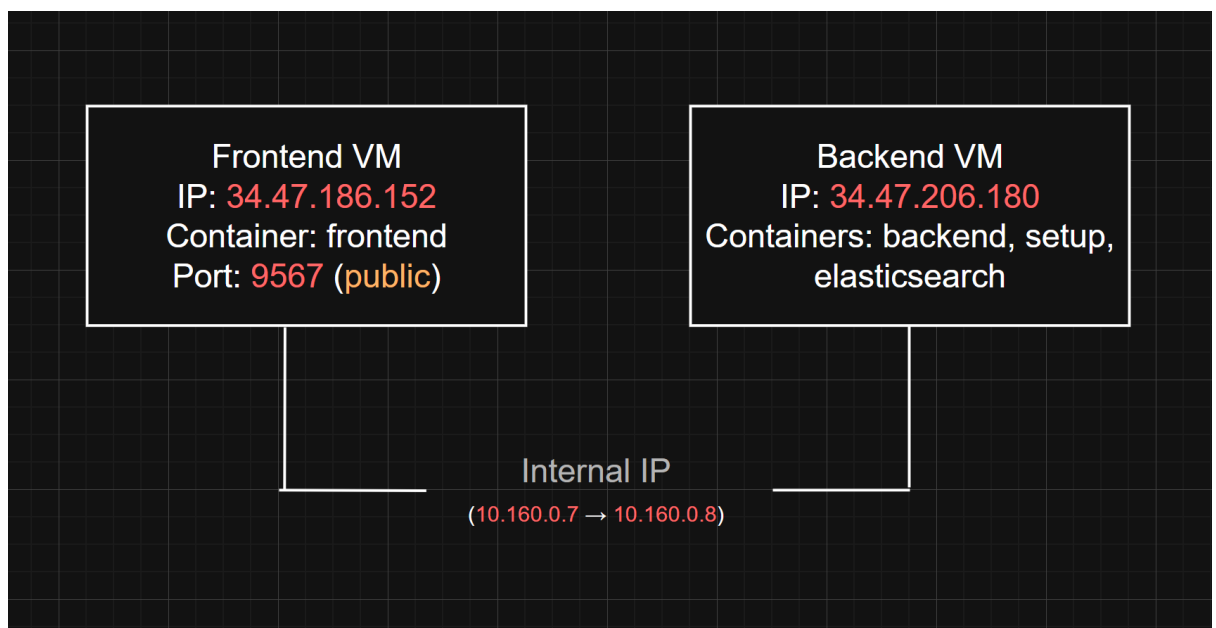
- fastapi: [adi362/fastapi | Docker Hub](#)
- elasticsearch: [adi362/elasticsearch general | Docker Hub](#)
- setup: [adi362/setup | Docker Hub](#)

Dockerfile URL:

<https://drive.google.com/drive/folders/18eJoLZKfgNLAzVrLIYCT9qDmpgTlwg5?usp=sharing>

◆ Overview of the Assignemnt:

This assignment involves deploying a multi-container Dockerized application using FastAPI and Elasticsearch across two separate virtual machines on GCP. The frontend must be public and isolated in a different VM, while the backend and Elasticsearch reside together on another VM.



✓ VM Instance Creation

1. Creating a VM instance for the frontend container (frontend-vm) on Google Cloud Platform.

Free trial status: ₹26,145.64 credit and 88 days remaining. Activate your full account to get unlimited access to all of Google Cloud—use any remaining credits, then pay only for what you use.

Google Cloud Docker-trial Search (/) for resources, docs, products, and more

Create an instance Create VM from... Equivalent code

Machine configuration

Name * frontend-vm

Region * asia-south1 (Mumbai) Zone * Any

Region is permanent Google will choose a zone on your behalf, maximizing VM obtainability. Zone is permanent.

☒ General purpose ☐ Compute optimized ☐ Memory optimized ☐ Storage optimized ☐ GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	CPU Platform
☐ C4	Consistently high performance	2 - 192	4 - 1,488 GB	Intel Emerald
☐ C4A	Arm-based consistently high performance	1 - 72	2 - 576 GB	Google Axion
☐ N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald
☐ C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire
☐ C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒ E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwell
☐ N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade
☐ N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan

Monthly estimate

\$44.58

That's about \$0.06 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 2 GB memory	\$43.38
10 GB balanced persistent disk	\$1.20
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$44.58

[Compute Engine pricing](#) [Cloud Operations pricing](#) [Less](#)

Create Cancel Equivalent code

2. Configuring machine type (E2), boot disk (Ubuntu), and firewall settings during VM creation.

Free trial status: ₹26,145.64 credit and 88 days remaining. Activate your full account to get unlimited access to all of Google Cloud—use any remaining credits, then pay only for what you use.

Google Cloud Docker-trial Search (/) for resources, docs, products, and more

Create an instance Create VM from... Equivalent code

Machine configuration

e2-custom-2-2048, asia-south1

OS and storage

Ubuntu 20.04 LTS

Data protection

Snapshot schedules

Networking

2 firewall rules, 1 network interface

Observability

Install Ops Agent

Security

Advanced

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

☐ Preset ☒ Custom

Creating a custom machine incurs additional costs

Cores

2 32 2 vCPU (1 core)

☐ Shared core

Memory

1 16 2 GB

[Advanced configurations](#)

Monthly estimate

\$44.58

That's about \$0.06 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
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[Compute Engine pricing](#) [Cloud Operations pricing](#) [Less](#)

Create Cancel Equivalent code

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Google Cloud Docker-trial Search (/) for resources, docs, products, and more

Create an instance Create VM from...

Machine configuration e2-custom-2-2048, asia-south1

OS and storage Ubuntu 20.04 LTS

Data protection Snapshot schedules

Networking 2 firewall rules, 1 network interface

Observability Install Ops Agent

Security

Advanced

Networking

Firewall ⓘ
Add tags and firewall rules to allow specific network traffic from the Internet

- ☒ Allow HTTP traffic
- ☒ Allow HTTPS traffic
- ☐ Allow Load Balancer Health Checks

Network tags ⓘ

Hostname ⓘ
Set a custom hostname for this instance or leave it default. Choice is permanent

IP forwarding ⓘ
☒ Enable

Network performance configuration

Network bandwidth ⓘ
☐ Enable per VM Tier_1 networking performance
Maximum outbound network bandwidth: 4Gbps
VM to Public IP: 4Gbps

Create Cancel Equivalent code

Monthly estimate

\$44.58
That's about \$0.06 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 2 GB memory	\$43.38
10 GB balanced persistent disk	\$1.20
Logging	Cost varies ⓘ
Monitoring	Cost varies ⓘ
Snapshot schedule	Cost varies ⓘ
Total	\$44.58

[Compute Engine pricing ⓘ](#)
[Cloud Operations pricing ⓘ](#)
[Less](#)

3. Creating a VM instance for the backend and Elasticsearch services (backend-vm) on GCP.

Free trial status: ₹26,145.64 credit and 88 days remaining. Activate your full account to get unlimited access to all of Google Cloud—use any remaining credits, then pay only for what you use.

Google Cloud Docker-trial Search (/) for resources, docs, products, and more

Create an instance Create VM from...

Machine configuration e2-medium, asia-south1

OS and storage Ubuntu 20.04 LTS

Data protection Snapshot schedules

Networking 2 firewall rules, 1 network interface

Observability Install Ops Agent

Security

Advanced

Machine configuration

Name *
backend-vm ⓘ

Region *
asia-south1 (Mumbai) ⓘ
Region is permanent

Zone *
Any ⓘ
Google will choose a zone on your behalf, maximizing VM obtainability. Zone is permanent.

☒ General purpose ☐ Compute optimized ☐ Memory optimized ☐ Storage optimized ☐ GPUs

Machine types for common workloads, optimized for cost and flexibility

Series ⓘ	Description	vCPUs ⓘ	Memory ⓘ	CPU Platform
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☐ N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade
☐ N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan

Create Cancel Equivalent code

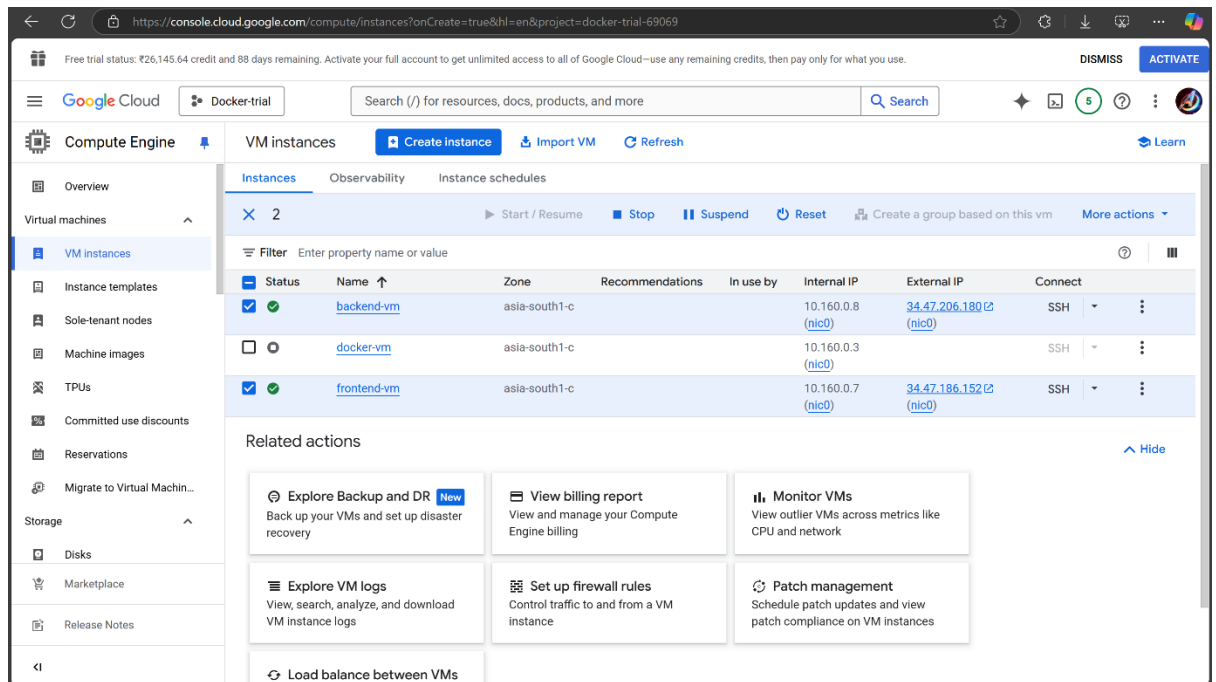
Monthly estimate

\$30.58
That's about \$0.04 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$29.38
10 GB balanced persistent disk	\$1.20
Logging	Cost varies ⓘ
Monitoring	Cost varies ⓘ
Snapshot schedule	Cost varies ⓘ
Total	\$30.58

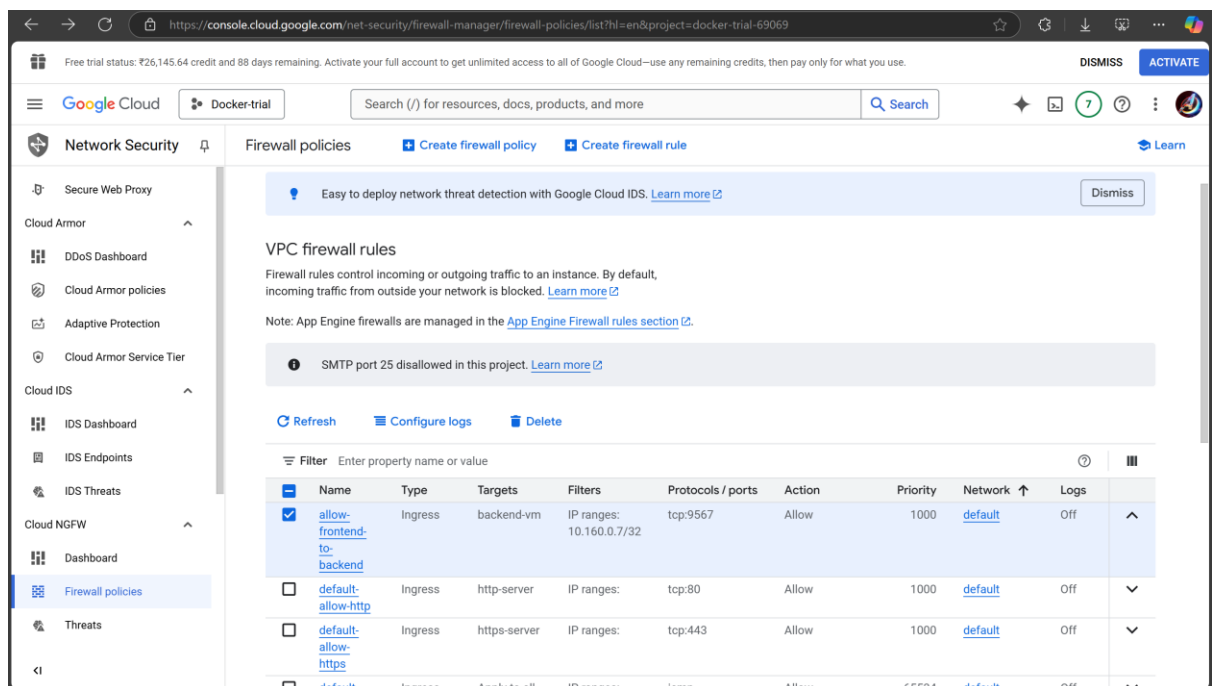
[Compute Engine pricing ⓘ](#)
[Cloud Operations pricing ⓘ](#)
[Less](#)

4. Two VM instances created

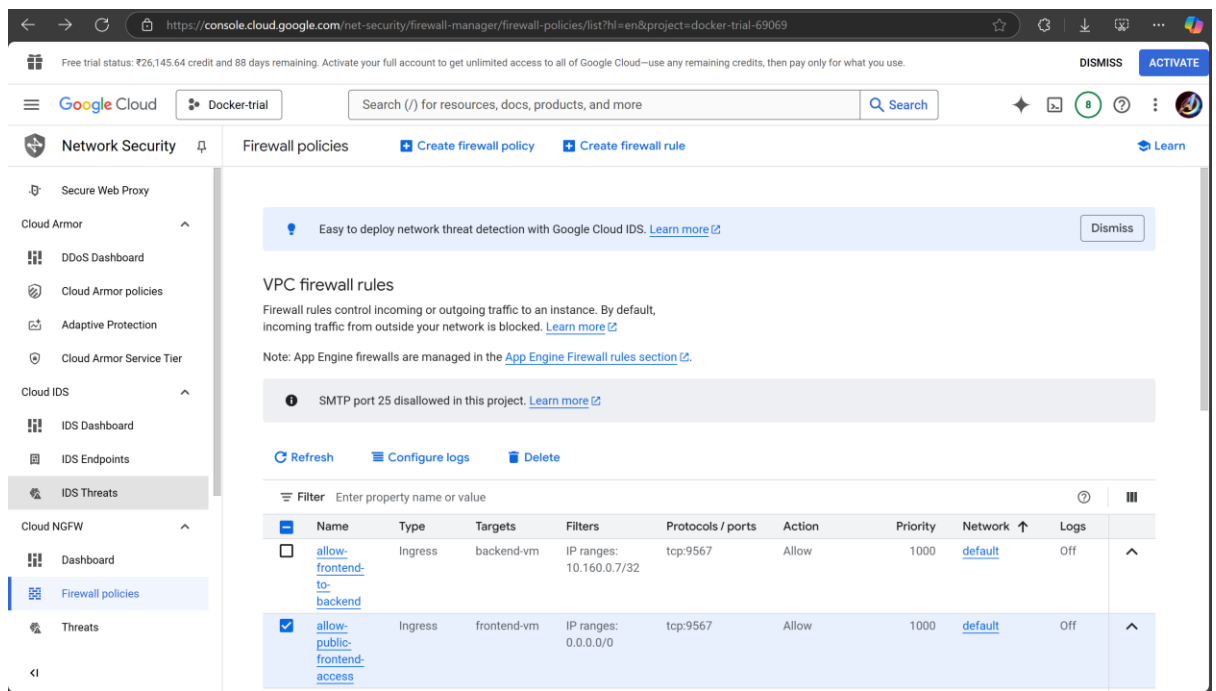


✓ Adding 2 Firewall Rules

1. Created a custom firewall rule to restrict backend access to only the frontend VM

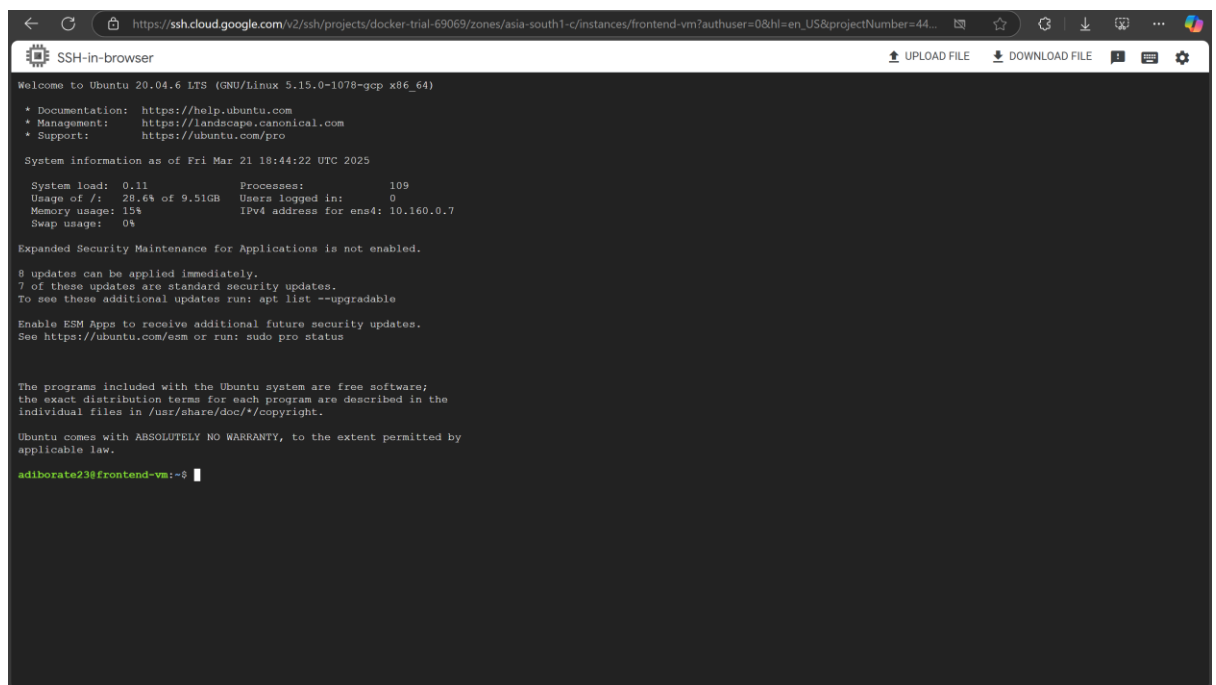


2. Created another custom firewall rule to allow public frontend access



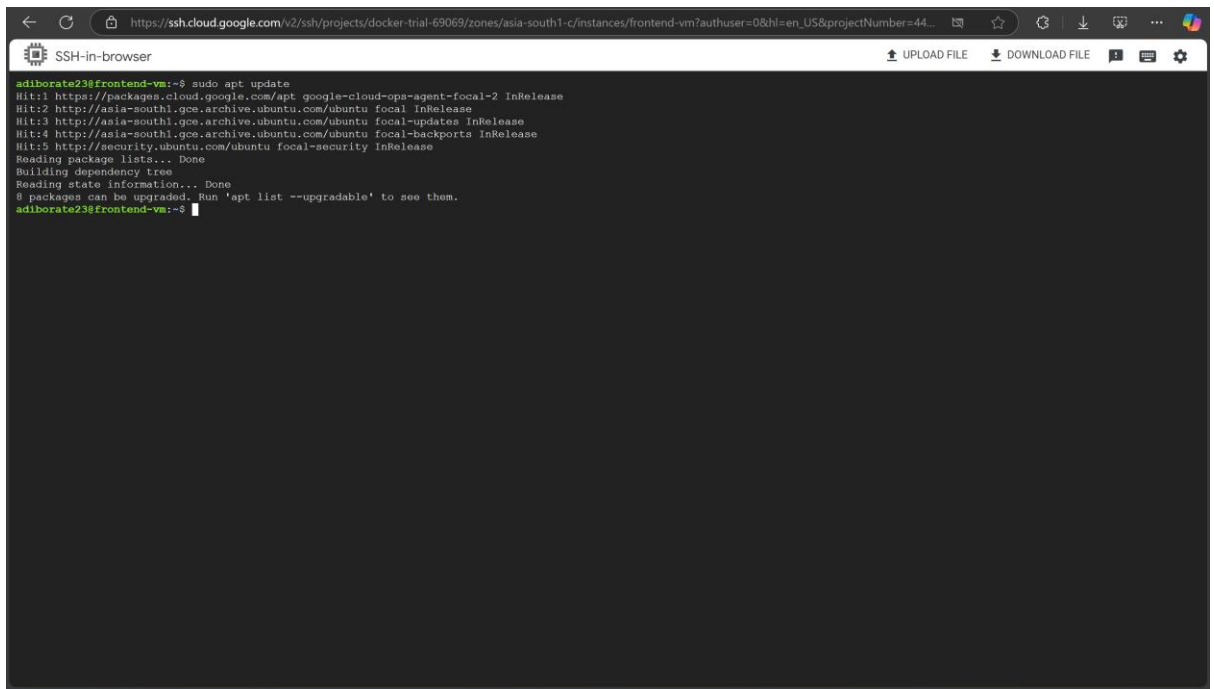
✓ Running frontend-vm by connecting SSH

1. Initial stage:



2. Installing Docker:

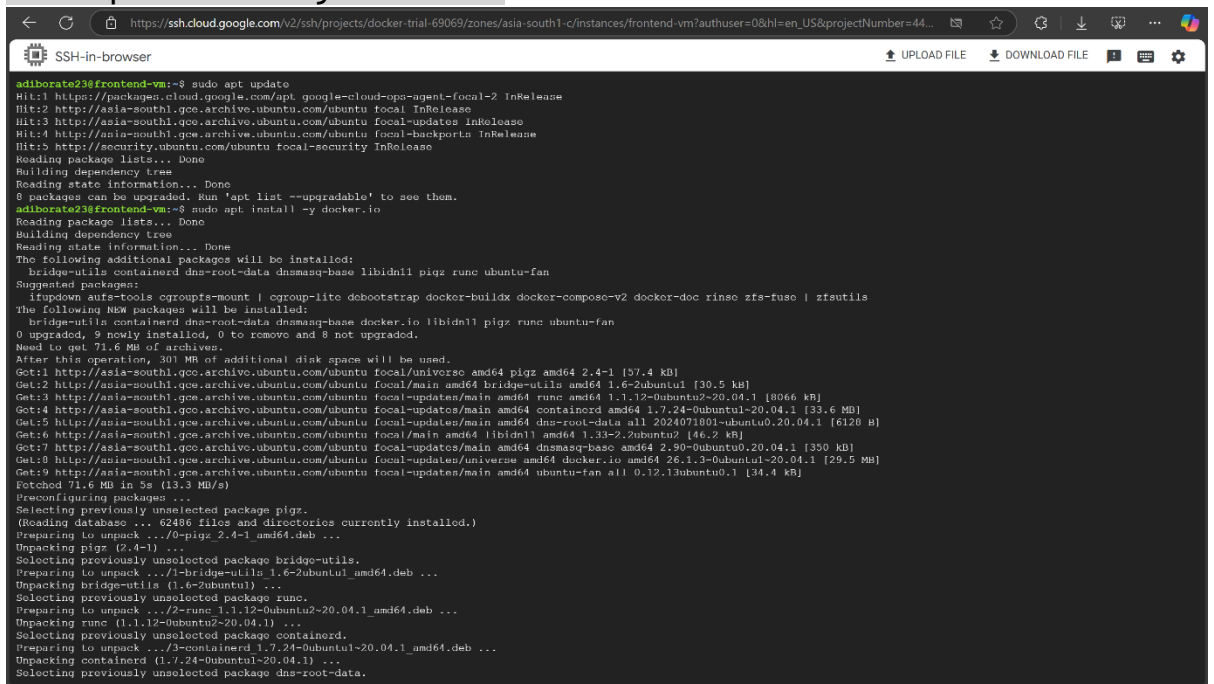
```
# Update package list
sudo apt update
```



```
adiborate23@frontend-vm:~$ sudo apt update
Hit:1 https://packages.cloud.google.com/apt google-cloud-ops-agent-focal-2 InRelease
Hit:2 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal InRelease
Hit:3 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:4 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
adiborate23@frontend-vm:~$
```

Install Docker

```
sudo apt install -y docker.io
```



```
adiborate23@frontend-vm:~$ sudo apt update
Hit:1 https://packages.cloud.google.com/apt google-cloud-ops-agent-focal-2 InRelease
Hit:2 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal InRelease
Hit:3 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:4 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
adiborate23@frontend-vm:~$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base libidn11 pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debotstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io libidn11 pigz runc ubuntu-fan
0 upgraded, 9 newly installed, 0 to remove and 8 not upgraded.
Need to get 71.6 MB of archives.
After this operation, 301 MB of additional disk space will be used.
Get:1 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal/universe amd64 pigz amd64 2.4-1 [57.4 kB]
Get:2 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal/main amd64 bridge-utils amd64 1.6-2ubuntu1 [30.5 kB]
Get:3 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates/main amd64 runc amd64 1.1.12-0ubuntu2-20.04.1 [8066 kB]
Get:4 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates/main amd64 containerd amd64 1.7.24-0ubuntu1-20.04.1 [33.6 MB]
Get:5 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates/main amd64 dns-root-data all 2024071801-ubuntu0.20.04.1 [6128 B]
Get:6 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal/main amd64 libidn11 amd64 1.32-2.2ubuntu2 [46.2 kB]
Get:7 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates/main amd64 dnsmasq-base amd64 2.90-0ubuntu0.20.04.1 [350 kB]
Get:8 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates/universe amd64 docker.io amd64 26.1.3-0ubuntu1-20.04.1 [29.5 MB]
Get:9 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates/main amd64 ubuntu-fan all 0.12.13ubuntu0.1 [34.4 kB]
Fetched 71.6 MB in 5s (13.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 62486 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.4-1_amd64.deb ...
Unpacking pigz (2.4-1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.6-2ubuntu1_amd64.deb ...
Unpacking bridge-utils (1.6-2ubuntu1) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.1.12-0ubuntu2-20.04.1_amd64.deb ...
Unpacking runc (1.1.12-0ubuntu2-20.04.1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_1.7.24-0ubuntu1-20.04.1_amd64.deb ...
Unpacking containerd (1.7.24-0ubuntu1-20.04.1) ...
Selecting previously unselected package dns-root-data.
Preparing to unpack .../4-dns-root-data_2024071801-ubuntu0.20.04.1_all.deb ...
Unpacking dns-root-data (2024071801-ubuntu0.20.04.1) ...
Selecting previously unselected package dnsmasq-base.
Preparing to unpack .../5-dnsmasq-base_2.90-0ubuntu0.20.04.1_amd64.deb ...
Unpacking dnsmasq-base (2.90-0ubuntu0.20.04.1) ...
Selecting previously unselected package libidn11.
Preparing to unpack .../6-libidn11_1.32-2.2ubuntu2_amd64.deb ...
Unpacking libidn11 (1.32-2.2ubuntu2) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../7-docker.io_26.1.3-0ubuntu1-20.04.1_amd64.deb ...
Unpacking docker.io (26.1.3-0ubuntu1-20.04.1) ...
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../8-ubuntu-fan_0.12.13ubuntu0.1_all.deb ...
Unpacking ubuntu-fan (0.12.13ubuntu0.1) ...
Setting up pigz (2.4-1) ...
Setting up bridge-utils (1.6-2ubuntu1) ...
Setting up runc (1.1.12-0ubuntu2-20.04.1) ...
Setting up containerd (1.7.24-0ubuntu1-20.04.1) ...
Setting up dns-root-data (2024071801-ubuntu0.20.04.1) ...
Setting up dnsmasq-base (2.90-0ubuntu0.20.04.1) ...
Setting up libidn11:amd64 (1.32-2.2ubuntu2) ...
Setting up docker.io (26.1.3-0ubuntu1-20.04.1) ...
Setting up ubuntu-fan (0.12.13ubuntu0.1) ...
```

Enable Docker service

```
sudo systemctl enable docker
```

Start Docker service

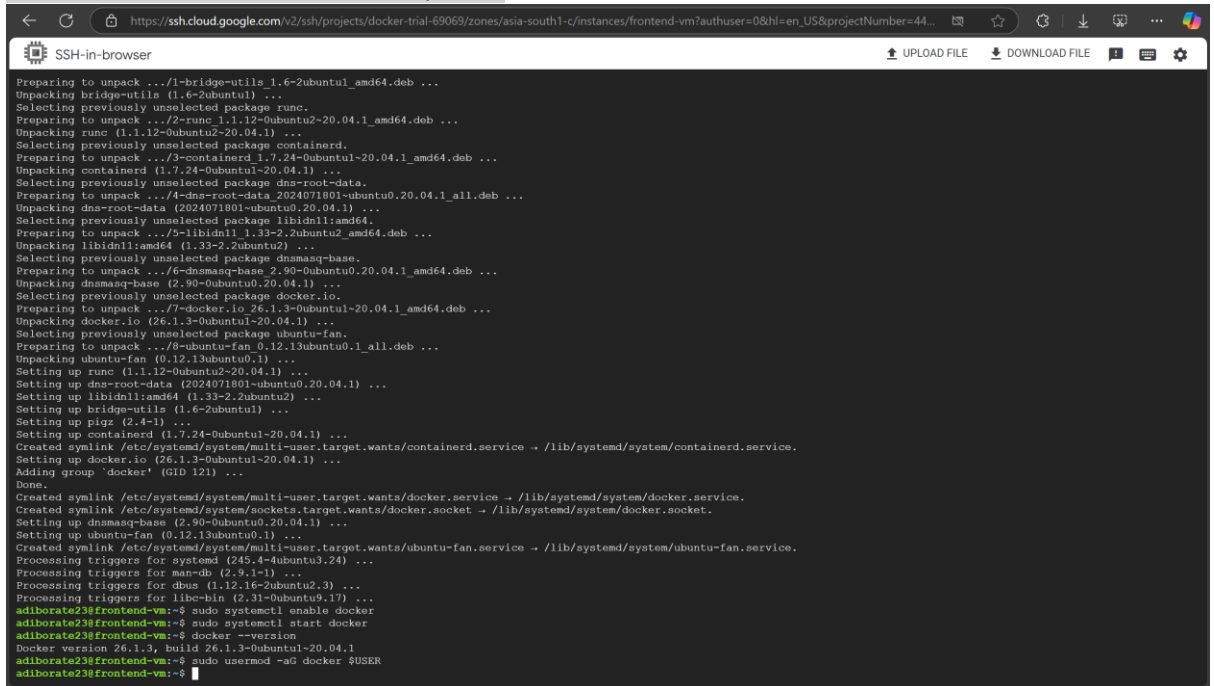
```
sudo systemctl start docker
```

Verify Docker installation

```
docker --version
```

#Add Your User to the Docker Group

```
sudo usermod -aG docker $USER
```

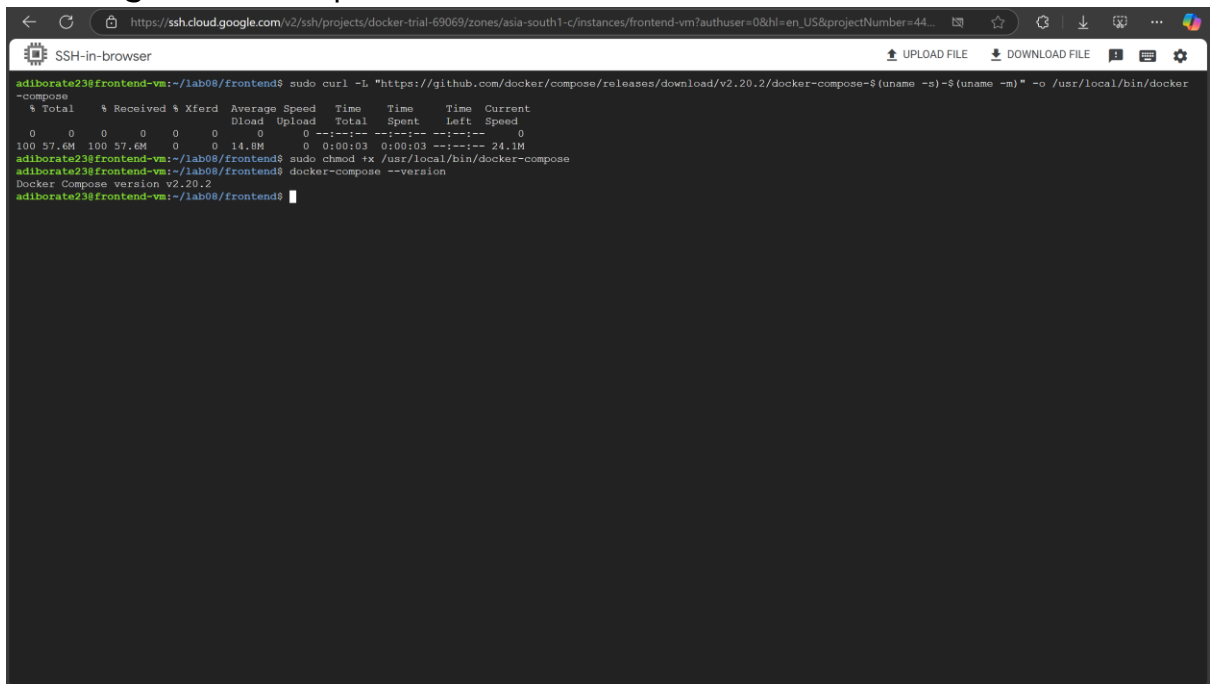


```

Preparing to unpack .../1-bridge-utils_1.6-2ubuntu1_amd64.deb ...
Unpacking bridge-utils (1.6-2ubuntu1) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.1.12-0ubuntu2-20.04.1_amd64.deb ...
Unpacking runc (1.1.12-0ubuntu2-20.04.1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_1.7.24-0ubuntu1-20.04.1_amd64.deb ...
Unpacking containerd (1.7.24-0ubuntu1-20.04.1) ...
Selecting previously unselected package dns-root-data.
Preparing to unpack .../4-dns-root-data_2024071801-ubuntu0.20.04.1_all.deb ...
Unpacking dns-root-data (2024071801-ubuntu0.20.04.1) ...
Selecting previously unselected package libidn11:amd64.
Preparing to unpack .../5-libidn11_1.33-2.2ubuntu2_amd64.deb ...
Unpacking libidn11:amd64 (1.33-2.2ubuntu2) ...
Selecting previously unselected package dnsmasq-base.
Preparing to unpack .../6-dnsmasq-base_2.90-0ubuntu0.20.04.1_amd64.deb ...
Unpacking dnsmasq-base (2.90-0ubuntu0.20.04.1) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../7-docker.io_26.1.3-0ubuntu1-20.04.1_amd64.deb ...
Unpacking docker.io (26.1.3-0ubuntu1-20.04.1) ...
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../8-ubuntu-fan_0.12.13ubuntu0.1_all.deb ...
Unpacking ubuntu-fan (0.12.13ubuntu0.1) ...
Setting up runc (1.1.12-0ubuntu2-20.04.1) ...
Setting up dns-root-data (2024071801-ubuntu0.20.04.1) ...
Setting up libidn11:amd64 (1.33-2.2ubuntu2) ...
Setting up bridge-utils (1.6-2ubuntu1) ...
Setting up pigz (2.4-1) ...
Setting up containerd (1.7.24-0ubuntu1-20.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /lib/systemd/system/containerd.service.
Setting up docker.io (26.1.3-0ubuntu1-20.04.1) ...
Adding group 'docker' (GID 121) ...
Done.
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /lib/systemd/system/docker.socket.
Setting up dnsmasq-base (2.90-0ubuntu0.20.04.1) ...
Setting up ubuntu-fan (0.12.13ubuntu0.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /lib/systemd/system/ubuntu-fan.service.
Processing triggers for systemd (245.4-4ubuntu3.24) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for dbus (1.12.16-2ubuntu2.3) ...
Processing triggers for libc-bin (2.31-0ubuntu9.17) ...
adiborate23@frontend-vm:~$ sudo systemctl enable docker
adiborate23@frontend-vm:~$ sudo systemctl start docker
adiborate23@frontend-vm:~$ docker --version
Docker version 26.1.3, build 26.1.3-0ubuntu1-20.04.1
adiborate23@frontend-vm:~$ sudo usermod -aG docker $USER
adiborate23@frontend-vm:~$

```

3. Installing docker-compose



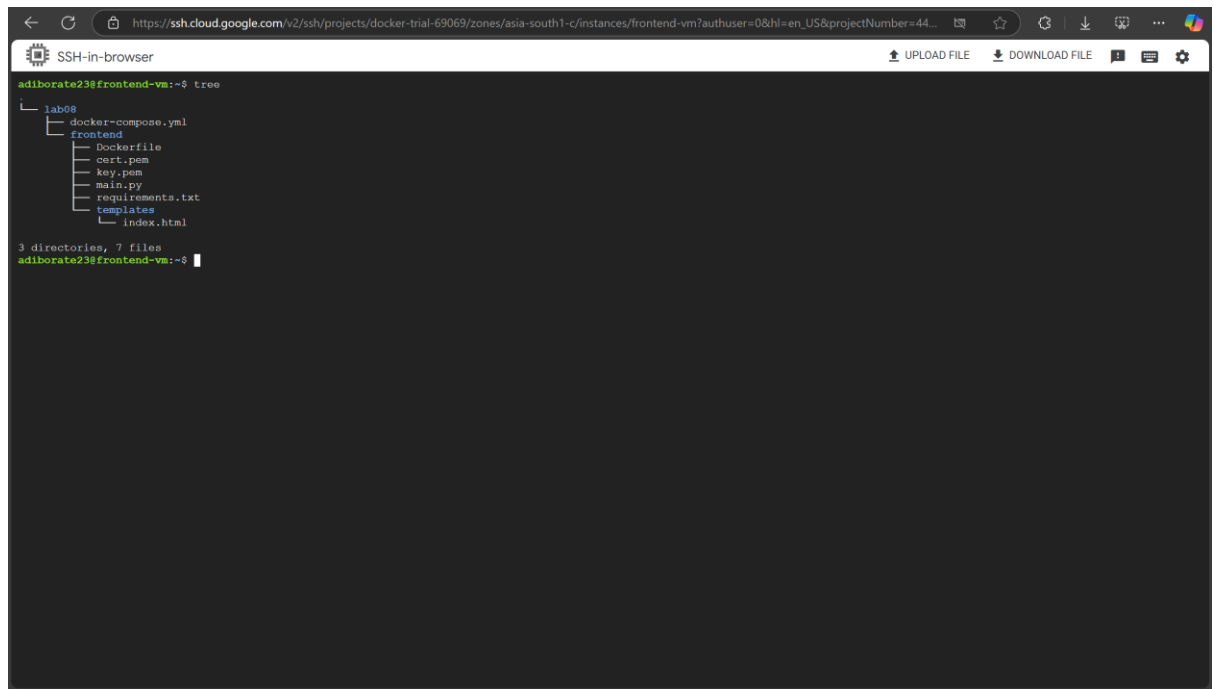
```

adiborate23@frontend-vm:~/lab08/frontend$ sudo curl -L "https://github.com/docker/compose/releases/download/v2.20.2/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker
-ccompose
% Total    % Received % Xferd Average Speed Time Time Time Current
           Dload Upload Total Spent Left Speed
100 57.6M 100 57.6M 0 0 14.8M 0 0:00:03 0:00:03 --:--:-- 24.1M
adiborate23@frontend-vm:~/lab08/frontend$ sudo chmod +x /usr/local/bin/docker-compose
adiborate23@frontend-vm:~/lab08/frontend$ docker-compose --version
Docker Compose version v2.20.2
adiborate23@frontend-vm:~/lab08/frontend$

```

✓ Setting Up Frontend

1. After writing necessary frontend files, this is how the directory structure is:

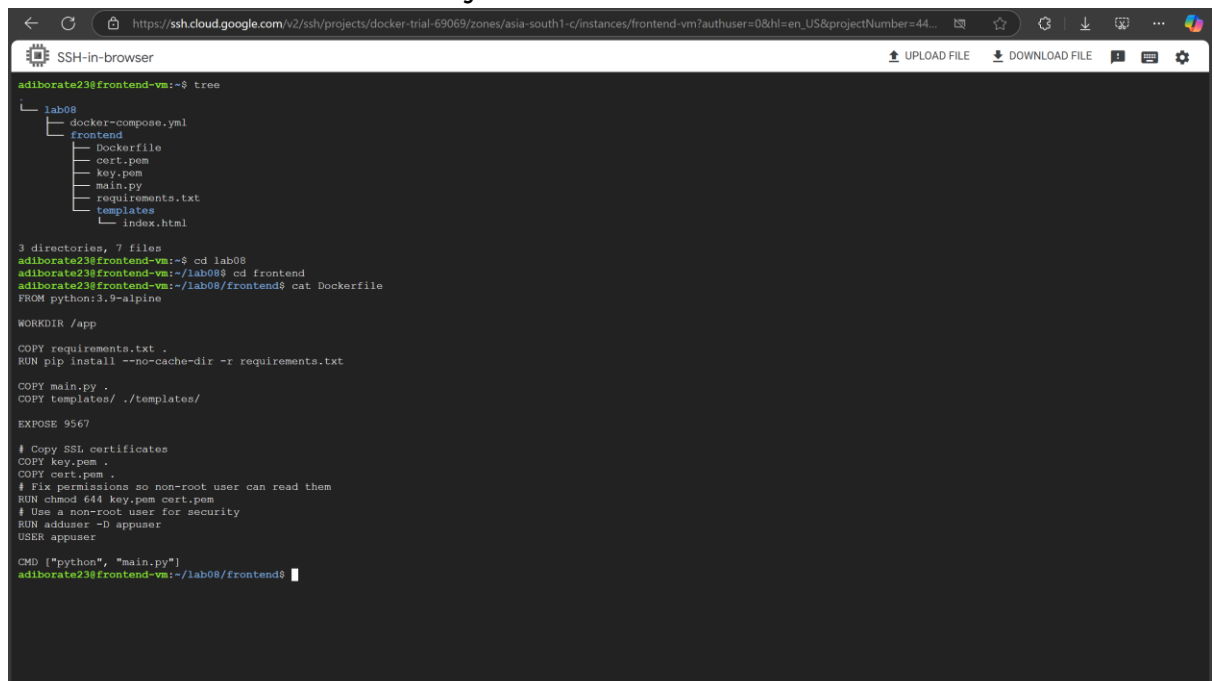


```
adiborate23@frontend-vm:~$ tree
.
├── lab08
│   ├── docker-compose.yml
│   └── frontend
│       ├── Dockerfile
│       ├── cert.pem
│       ├── key.pem
│       ├── main.py
│       ├── requirements.txt
│       ├── templates
│       └── index.html
3 directories, 7 files
adiborate23@frontend-vm:~$
```

FastAPI, by default, works with HTTP and requires configuration to use HTTPS. The cert.pem and key.pem are the certificate files for setting up HTTPS access, generated with this command:

```
openssl req -x509 -newkey rsa:4096 -nodes -out cert.pem -keyout key.pem -days 365
```

2. Dockerfile in frontend directory:



```
adiborate23@frontend-vm:~$ tree
.
├── lab08
│   ├── docker-compose.yml
│   └── frontend
│       ├── Dockerfile
│       ├── cert.pem
│       ├── key.pem
│       ├── main.py
│       ├── requirements.txt
│       ├── templates
│       └── index.html
3 directories, 7 files
adiborate23@frontend-vm:~$ cd lab08
adiborate23@frontend-vm:~/lab08$ cd frontend
adiborate23@frontend-vm:~/lab08/frontend$ cat Dockerfile
FROM python:3.9-alpine

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY main.py .
COPY templates/ ./templates/

EXPOSE 9567

# Copy SSL certificates
COPY key.pem .
COPY cert.pem .
# Fix permissions so non-root user can read them
RUN chmod 644 key.pem cert.pem
# Use a non-root user for security
RUN adduser -D appuser
USER appuser

CMD ["python", "main.py"]
adiborate23@frontend-vm:~/lab08/frontend$
```

(To view entire codebase, please refer to the attached GitHub Repository URL at the top)

✅ Build and start the frontend container

1. `docker-compose up -d` : builds (if needed) and starts the frontend container in detached mode, setting it up with the defined ports, volumes, and environment as specified in `docker-compose.yml`

```

$ Total      % Received % Xferd Average Speed      Time      Time      Time Current
   Dload    Upload   Total           Dload    Upload               Speed
100 57.6M 100 57.6M    0    0 14.8M    0 0:00:03 0:00:03 --:--:-- 24.1M
adiborate23@frontend-vm:~/lab08/frontend$ sudo chmod +x /usr/local/bin/docker-compose
adiborate23@frontend-vm:~/lab08/frontend$ docker-compose --version
Docker Compose version v2.20.2
adiborate23@frontend-vm:~/lab08/frontend$ docker-compose up -d
[*] Running 1/1
! frontend Warning
[*] Building 18.0s (15/15) FINISHED
-- [frontend internal] load build definition from Dockerfile
--> transferring dockerfile: 444B
-- [frontend internal] load metadata for docker.io/library/python:3.9-alpine
-- [frontend internal] load .dockerignore
--> transferring context: 2B
-- [frontend 1/10] FROM docker.io/library/python:3.9-alpine@sha256:e345f1410de8c8c40a0afac784deabce796a52f26965c41296a71044fb47fab
--> resolve docker.io/library/python:3.9-alpine@sha256:e345f1410de8c8c40a0afac784deabce796a52f26965c41296a71044fb47fab
--> sha256:e345f1410de8c8c40a0afac784deabce796a52f26965c41296a71044fb47fab 10.29KB / 10.29KB
--> sha256:1f0c320daabff3222a6e42480287f39907a5c32187acdd1b0e639095444c 1.73KB / 1.73KB
--> sha256:1f0c320daabff3222a6e42480287f39907a5c32187acdd1b0e639095444c 5.07KB / 5.07KB
--> sha256:1f0c32174bc91741f4f34a96d85011092101a032a93a388b7ee96e6c2d5c870 3.64MB / 3.64MB
--> sha256:6e0fc7c91101bd06c8dfbc20c0cf770af96b0168c83323ab053c10545537a5be 450.62KB / 450.62KB
--> sha256:ca9948a85d045a6a2a32794cfac4ba5d9f271f65aed76114458b1b124f4d4ea3 14.07MB / 14.07MB
--> extracting sha256:1f0c32174bc91741f4f34a96d85011092101a032a93a388b7ee96e6c2d5c870
--> extracting sha256:6e0fc7c91101bd06c8dfbc20c0cf770af96b0168c83323ab053c10545537a5be
--> extracting sha256:ca9948a85d045a6a2a32794cfac4ba5d9f271f65aed76114458b1b124f4d4ea3
--> extracting sha256:ca32370b5d96d783ff338254085ca725fb5d3c1a9603100dc474063c4aaf192
[frontend internal] load build context
--> transferring context: 9.13KB
-- [frontend 2/10] WORKDIR /app
-- [frontend 3/10] COPY requirements.txt .
-- [frontend 4/10] RUN pip install --no-cache-dir -r requirements.txt
-- [frontend 5/10] COPY main.py .
-- [frontend 6/10] COPY templates/ ./templates/
-- [frontend 7/10] COPY key.pem .
-- [frontend 8/10] COPY cert.pem .
-- [frontend 9/10] RUN chmod 644 key.pem cert.pem
-- [frontend 10/10] RUN adduser -D appuser
-- [frontend] exporting to image
--> exporting layers
--> writing image sha256:f3fa5eb0b66e207bfa473973d85616a55dbf637a4a1644a33427a1c137bf
--> naming to docker.io/library/fastapi:1
[*] Running 2/2
✓ Network lab08 default Created
✓ Container lab08-frontend-1 Started
adiborate23@frontend-vm:~/lab08/frontend$

```

2. `docker ps` : shows all running containers

```

adiborate23@frontend-vm:~/lab08/frontend$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
3d67858f31d7   fastapi:1 "python main.py"        3 minutes ago    Up 3 minutes    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp    lab08-frontend-1
adiborate23@frontend-vm:~/lab08/frontend$

```

3. Test frontend endpoint with curl (internal check)

```

adiborate23@frontend-vm:~/lab08/frontend$ curl -k https://localhost:9567
<!DOCTYPE html>
<html>
<head>
<title>Elasticsearch Interface</title>
</head>
<body>
<h1>Elasticsearch Document Manager</h1>
<h2>Search Document</h2>
<form action="/get" method="post">
  <input type="text" name="query" placeholder="Enter search query" required>
  <button type="submit">Get</button>
</form>
<h2>Insert Document</h2>
<form action="/insert" method="post">
  <textarea name="content" rows="5" cols="50" placeholder="Enter document content" required></textarea><br>
  <button type="submit">insert</button>
</form>
<div id="result" style="margin-top: 20px;">
  <h2>Result:</h2>
  <p>No results to display.</p>
</div>
</body>
</html>adiborate23@frontend-vm:~/lab08/frontend$

```

4. App is bound to `0.0.0.0` in code to allow external access using this part of code in `main.py`

```
# Run the FastAPI app with uvicorn on port 9567, binding to all addresses (0.0.0.0)
if __name__ == "__main__":
    uvicorn.run(
        app,
        host="0.0.0.0",
        port=9567,
        ssl_keyfile="key.pem",
        ssl_certfile="cert.pem"
    )
```

5. Checking listening ports using netstat -antp | grep LISTEN

```
adiborate23@frontend-vm:~/lab08/frontend$ netstat -antp | grep LISTEN
Command 'netstat' not found, but can be installed with:

apt install net-tools

adiborate23@frontend-vm:~/lab08/frontend$ sudo apt update
Hit:1 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Hit:3 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-focal-2 InRelease
Hit:5 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal-backports InRelease
Fetched 128 kB in 1s (138 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
8 packages can be upgraded. Run 'apt list --upgradable' to see them.
adiborate23@frontend-vm:~/lab08/frontend$ sudo apt install net-tools
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
 net-tools
0 upgraded, 1 newly installed, 0 to remove and 8 not upgraded.
Need to get 196 kB of archives.
After this operation, 864 kB of additional disk space will be used.
Get:1 http://asia-south1.gce.archive.ubuntu.com/ubuntu focal/main amd64 net-tools amd64 1.60+git20180626.aebd88e-lubuntui [196 kB]
Fetched 196 kB in 1s (355 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 62860 files and directories currently installed.)
Preparing to unpack .../net-tools_1.60+git20180626.aebd88e-lubuntui_amd64.deb ...
Unpacking net-tools (1.60+git20180626.aebd88e-lubuntui) ...
Setting up net-tools (1.60+git20180626.aebd88e-lubuntui) ...
Processing triggers for man-db (2.9.1-1) ...
adiborate23@frontend-vm:~/lab08/frontend$ netstat -antp | grep LISTEN
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN     -
tcp        0      0 0.0.0.0:9567            0.0.0.0:*               LISTEN     -
tcp        0      0 0.0.0.0:20202           0.0.0.0:*               LISTEN     -
tcp        0      0 0.0.0.0:53:53          0.0.0.0:*               LISTEN     -
tcp6       0      0 :::22                  :::*                    LISTEN     -
tcp6       0      0 :::9567                 :::*                    LISTEN     -
tcp6       0      0 :::20201                :::*                    LISTEN     -
adiborate23@frontend-vm:~/lab08/frontend$
```

6. Screenshot of docker images

```
adiborate23@frontend-vm:~/lab08/frontend$ netstat -antp | grep LISTEN
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN     -
tcp        0      0 0.0.0.0:9567            0.0.0.0:*               LISTEN     -
tcp        0      0 0.0.0.0:20202           0.0.0.0:*               LISTEN     -
tcp        0      0 0.0.0.0:53:53          0.0.0.0:*               LISTEN     -
tcp6       0      0 :::22                  :::*                    LISTEN     -
tcp6       0      0 :::9567                 :::*                    LISTEN     -
tcp6       0      0 :::20201                :::*                    LISTEN     -
adiborate23@frontend-vm:~/lab08/frontend$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
fastapi 1 f3fe5eb6be66 23 minutes ago 70.7MB
adiborate23@frontend-vm:~/lab08/frontend$
```

7. docker inspect on frontend container

```
SSH-in-browser
adiborate23@frontend-vm:~$ docker inspect 3d67858f31d7
[
  {
    "Id": "3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869",
    "Created": "2025-03-22T05:20:46.677026827Z",
    "Path": "python",
    "Args": [
      "main.py"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 53904,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2025-03-22T05:20:47.219868357Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:f3fe5eb6be6e207bfa473973d8561ea55d5ebf637a4a1644a33427a161c37bf",
    "ResolveConfPath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/resolve.conf",
    "HostnamePath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/hostname",
    "HostsPath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/hosts",
    "LogPath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869-jso
n.log",
    "Name": "/lab08-frontend-1",
    "RestartCount": 0,
    "Driver": "overlay2",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "docker-default",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
        "Config": {}
      },
      "NetworkMode": "lab08_default",
      "PortBindings": {
        "9567/tcp": [
          {
            "HostIp": "",
            "HostPort": "9567"
          }
        ]
      },
      "RestartPolicy": {
        "Name": "no",
        "MaximumRetryCount": 0
      },
      "AutoRemove": false,
      "VolumeDriver": "",
      "VolumesFrom": null,
      "ConsoleSize": {
        0,
        0
      },
      "CapAdd": null,
      "CapDrop": null,
      "CgroupnsMode": "host",
      "Dns": null,
      "DnsOptions": null,
      "DnsSearch": null,
      "ExtraHosts": {},
      "GroupAdd": null,
      "IpcMode": "private",
      "Cgroup": "",
      "Links": null,
      "OomScoreAdj": 0,
      "PidMode": "",
      "Privileged": false,
      "PublishAllPorts": false,
      "ReadonlyRootfs": false,
      "SecurityOpt": null,
      "UTSMode": "",
      "UsernsMode": "",
      "ShmSize": 67108864,
      "Runtime": "runc",
      "Isolation": "",
      "CpuShares": 0,
      "Memory": 0,
      "NanoCpus": 0,
      "CgroupParent": "",
      "BlkioWeight": 0,
      "BlkioWeightDevice": null,
      "BlkioDeviceReadBps": null,
      "BlkioDeviceWriteBps": null,
      "BlkioDeviceReadIOps": null,
      "BlkioDeviceWriteIOps": null,
      "CpuPeriod": 0,
    }
  }
]
```

```
SSH-in-browser

"CPUQuota": 0,
"CPURealtimePeriod": 0,
"CPURealtimeRuntime": 0,
"CpusetCpus": "",
"CpusetMems": "",
"Devices": null,
"DeviceGroupRules": null,
"DeviceRequests": null,
"MemoryReservation": 0,
"MemorySwap": 0,
"MemorySwappiness": null,
"OOMKillDisable": false,
"PidsLimit": null,
"Ulimits": null,
"CpuCount": 0,
"CpuPercent": 0,
"IOMaximumOps": 0,
"IOMaximumBandwidth": 0,
"MaskedPaths": [
  "/proc/asound",
  "/proc/acpi",
  "/proc/kcore",
  "/proc/kmsg",
  "/proc/latency_stats",
  "/proc/timer_list",
  "/proc/timer_stats",
  "/proc/sched_debug",
  "/proc/scsi",
  "/sys/firmware",
  "/sys/devices/virtual/powercap"
],
"ReadonlyPaths": [
  "/proc/bus",
  "/proc/fs",
  "/proc/i915",
  "/proc/irq",
  "/proc/sys",
  "/proc/sysrq-trigger"
],
"GraphDriver": {
  "Data": {
    "LowerDir": [
      "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59ceff3f306ba402-init/diff:/var/lib/docker/overlay2/n21ptadgok7a9iqujxphhbbf/diff:/var/lib/docker/overlay2/sdnptey7mllzxt4065gtr8z3/diff:/var/lib/docker/overlay2/58f1nnq3t75gs19khgfkveb8t/diff:/var/lib/docker/overlay2/nkhdslvbf3k1joaw93jutrdrw/diff:/var/lib/docker/overlay2/asjrlimgshuze3kg9eslodo48/diff:/var/lib/docker/overlay2/inb827fql12cg47j7ivlda4pb/diff:/var/lib/docker/overlay2/rczyp21j590wkk53lvqbt9vbq/diff:/var/lib/docker/overlay2/K7a4rwkutdkxloj49k0h549/diff:/var/lib/docker/overlay2/svmvjxv49pfn0v367s1b44g/diff:/var/lib/docker/overlay2/a47a02bc96139d6e7300e03aed5e2501fdb7abdb258e008687825e4d8d9b46/diff:/var/lib/docker/overlay2/8856020e8cabfcce15e3f074259e190d15ec67c28e80bcb6a557822c518657/diff:/var/lib/docker/overlay2/elclbe07b7b249b739be9c56e8f26ed7a94db966396cd663423a4af238dac30b/diff:/var/lib/docker/overlay2/72a869f3bfc1d615d412d6ba4728218cb9716545ef2385fae921041d7400b356/diff",
      "/mergedDir"
    ],
    "MergedDir": [
      "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59ceff3f306ba402/merged"
    ]
  }
},
"UpperDir": [
  "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59ceff3f306ba402/diff",
  "/workDir"
],
"Name": "overlay2"
},
"Mounts": [
  {
    "Hostname": "3d67858f31d7",
    "Domainname": "",
    "User": "appuser",
    "AttachStdin": false,
    "AttachStdout": true,
    "AttachStderr": true,
    "ExposedPorts": {
      "9567/tcp": {}
    },
    "TTY": false,
    "OpenStdin": false,
    "StdinOnce": false,
    "Env": [
      "PATH=/usr/local/bin:/usr/local/sbin:/usr/bin:/sbin:/bin",
      "LANG=C.UTF-8",
      "GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
      "PYTHON_VERSION=3.9.21",
      "PYTHON_SHA256=3126f9592c9b0d798584755f2bf7b081falca35ce7a6fe980108d752a05bb1"
    ],
    "Cmd": [
      "python",
      "main.py"
    ],
    "Image": "fantapi1",
    "Volumes": null,
    "WorkingDir": "/app",
    "Entrypoint": null,
    "Onbuild": null,
    "Labels": {
      "com.docker.compose.config-hash": "cc01e2fa156de645eccc1efe80efcbfec1f887b7da6ec5bc02978ec74cb4bf7b",
      "com.docker.compose.container-number": "1",
      "com.docker.compose.depends_on": "",
      "com.docker.compose.image": "sha256:f3fe5eb6ba66e207bfa473973d85616a55d1abf637a4a1644a33427a161c37bf",
      "com.docker.compose.oneoff": "false",
      "com.docker.compose.project": "lab08",
      "com.docker.compose.project.config_files": "/home/adiborate23/lab08/docker-compose.yml",
      "com.docker.compose.project.working_dir": "/home/adiborate23/lab08",
      "com.docker.compose.service": "frontend",
      "com.docker.compose.version": "2.20.2"
    }
  }
]
```

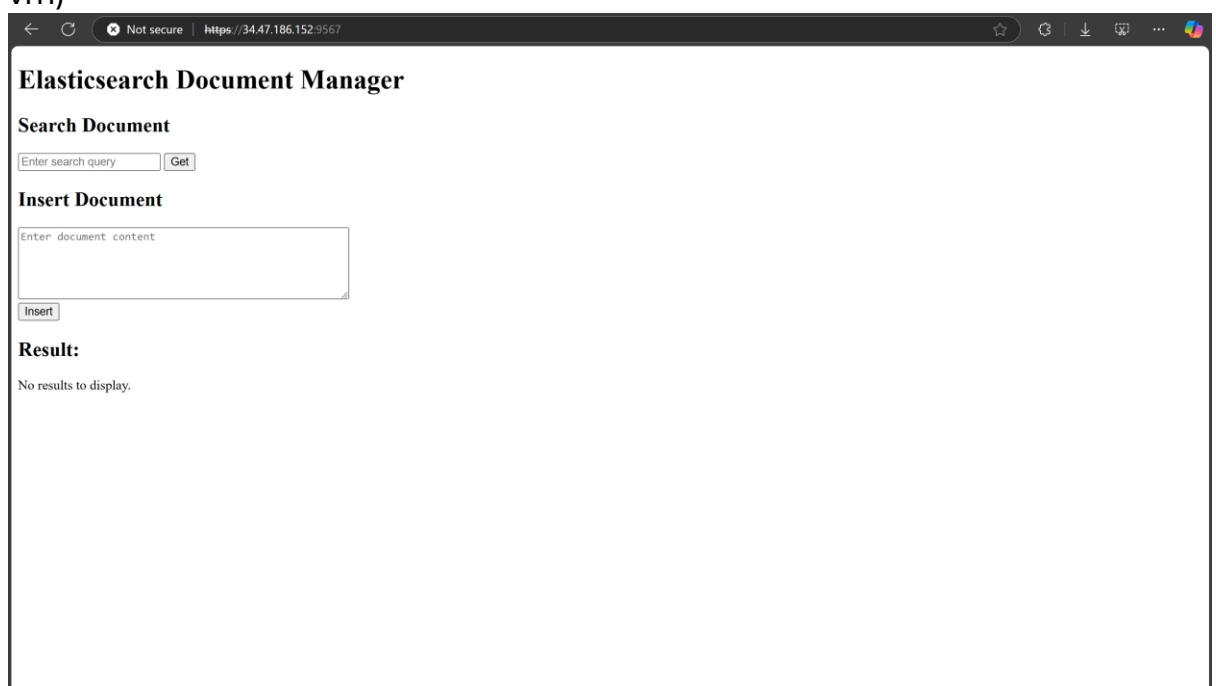


```

},
"NetworkSettings": {
  "Bridge": "",
  "SandboxIP": "a6e59060cfdb4ef47d4bb14c05c0bf80eeb5d14c5d6508738df52511aade0b1f",
  "SandboxKey": "/var/run/docker/netns/a6e59060cfdb",
  "Ports": {
    "9567/tcp": [
      {
        "HostIp": "0.0.0.0",
        "HostPort": "9567"
      },
      {
        "HostIp": "::",
        "HostPort": "9567"
      }
    ]
  }
},
"HairpinMode": false,
"LinkLocalIPv6Address": "",
"LinkLocalIPv6PrefixLen": 0,
"SecondaryIPAddresses": null,
"SecondaryIPv6Addresses": null,
"EndpointID": "",
"Gateway": "",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"IPAddress": "",
"IPPrefixLen": 0,
"IPv6Gateway": "",
"MacAddress": "",
"Networks": {
  "lab08 default": {
    "IPAMConfig": null,
    "Links": null,
    "Aliases": [
      "lab08-frontend-1",
      "frontend"
    ],
    "MacAddress": "02:42:ac:12:00:02",
    "NetworkID": "b9c4c7be07bc75db6f3d7a7c16b0521cfc6f6e040fe779c45f36faa4001afad9",
    "EndpointID": "937958223a69alc69d5e38602354fb656d77ccbde7b05862ce7bb0b813d8af67",
    "Gateway": "172.18.0.1",
    "IPAddress": "172.18.0.2",
    "IPPrefixLen": 16,
    "IPv6Gateway": "",
    "GlobalIPv6Address": "",
    "GlobalIPv6PrefixLen": 0,
    "DriverOpts": null
  }
},
"DNSNames": [
  "lab08-frontend-1",
  "frontend",
  "3d67858f31d7"
]
}
}
}
}
}
adiborate23@frontend-vm:~$

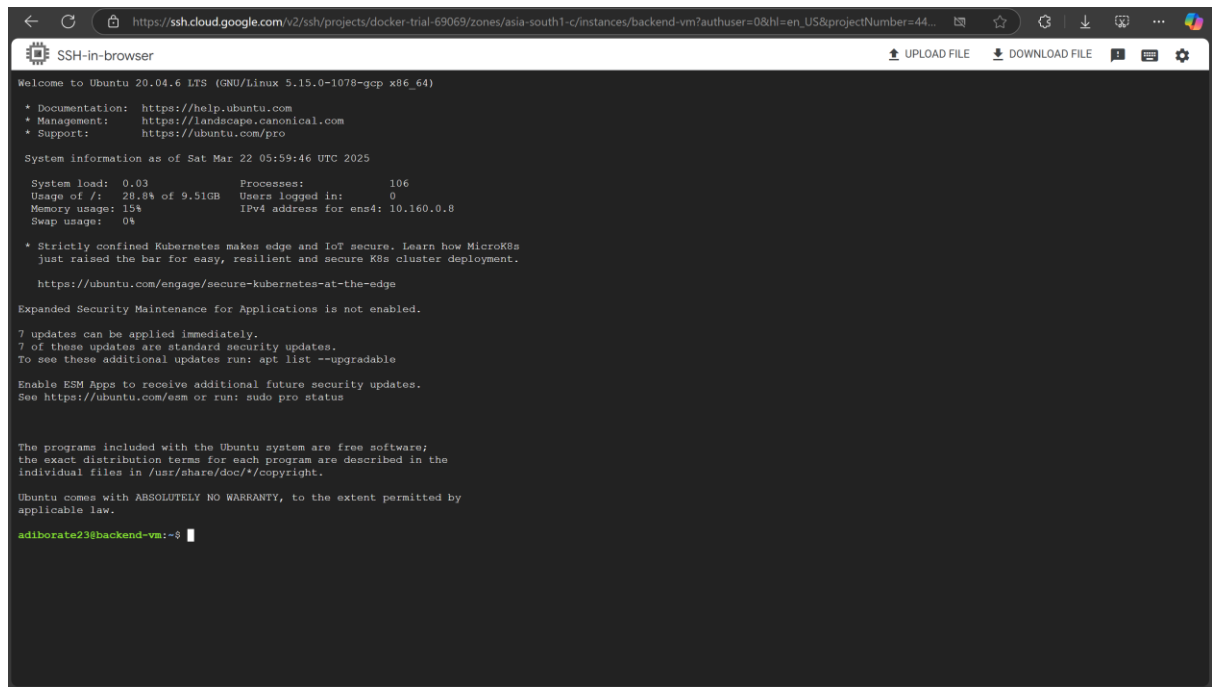
```

8. This is how the frontend looks like (accessed via external IP of frontend-vm)



✓ Setting Up Backend

1. Running backend-vm by connecting SSH (Initial Stage):



```

Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1078-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Mar 22 05:59:46 UTC 2025

System load:  0.03               Processes:    106
Usage of /:   28.8% of 9.51GB     Users logged in:  0
Memory usage: 15%               IPv4 address for ens4: 10.160.0.8
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
7 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

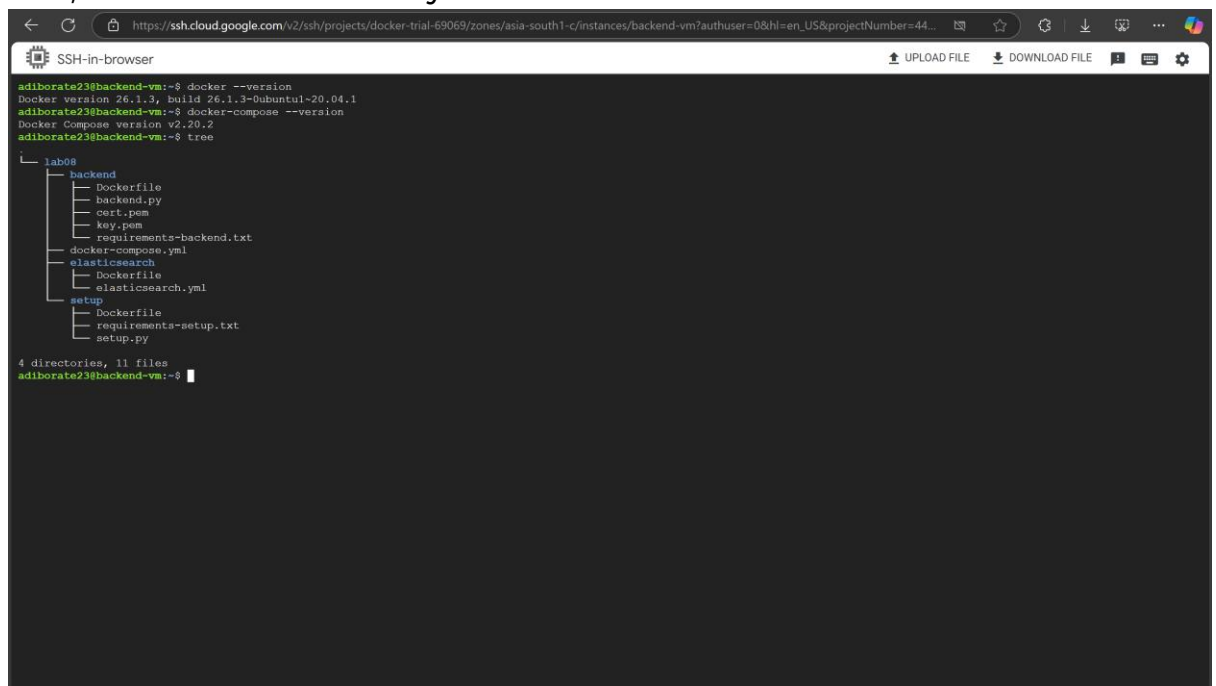
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

adiborate23@backend-vm:~$

```

2. After installing same required packages, and writing necessary backend files, this is how the directory structure is:



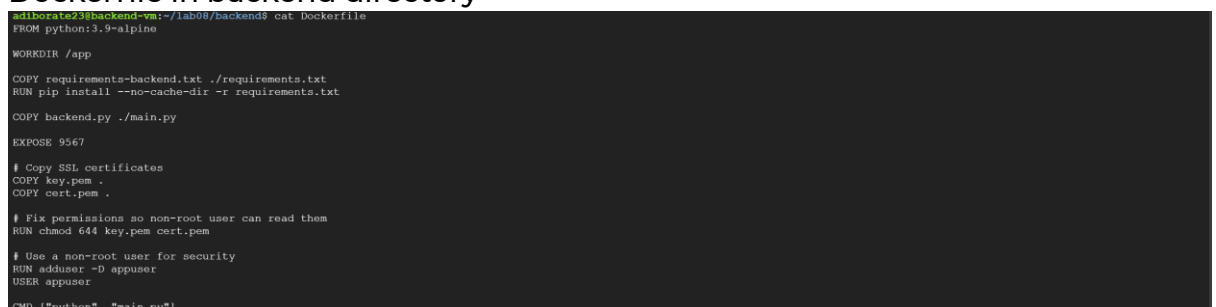
```

adiborate23@backend-vm:~$ docker --version
Docker version 26.1.3, build 26.1.3-0ubuntu1-20.04.1
adiborate23@backend-vm:~$ docker-compose --version
Docker Compose version v2.20.2
adiborate23@backend-vm:~$ tree
.
├── lab08
│   ├── backend
│   │   ├── Dockerfile
│   │   ├── backend.py
│   │   ├── cert.pem
│   │   ├── key.pem
│   │   └── requirements-backend.txt
│   ├── docker-compose.yml
│   ├── elasticsearch
│   │   ├── Dockerfile
│   │   └── elasticsearch.yml
│   └── setup
│       ├── Dockerfile
│       ├── requirements-setup.txt
│       └── setup.py
└── 4 directories, 11 files

adiborate23@backend-vm:~$

```

3. Dockerfile in backend directory



```

adiborate23@backend-vm:~/lab08/backend$ cat Dockerfile
FROM python:3.9-alpine

WORKDIR /app

COPY requirements-backend.txt ./requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

COPY backend.py ./main.py

EXPOSE 9567

# Copy SSL certificates
COPY key.pem .
COPY cert.pem .

# Fix permissions so non-root user can read them
RUN chmod 644 key.pem cert.pem

# Use a non-root user for security
RUN adduser -D appuser
USER appuser

CMD ["python", "main.py"]

```

```
adiborate23@backend-vm:~/lab08/elasticsearch$ cat Dockerfile
FROM elasticsearch:7.14.0

# Add configuration for persistence
COPY elasticsearch.yml /usr/share/elasticsearch/config/elasticsearch.yml

# Set permissions
USER root
RUN chown elasticsearch:elasticsearch /usr/share/elasticsearch/config/elasticsearch.yml
USER elasticsearch

# Expose the required port used by elasticsearch
EXPOSE 9200 9300
```

```
adiborate23@backend-vm:~/lab08/setup$ cat Dockerfile
FROM python:3.9-alpine

WORKDIR /app

COPY requirements-setup.txt ./requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

COPY setup.py ./setup.py

CMD ["python", "setup.py"]
```

✓ Building and Running the containers

```
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south-1/c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...
SSH-in-browser
[+] Building 3.8s (29/29) FINISHED
=> [elasticsearch internal] load build definition from Dockerfile
=> -- transferring Dockerfile: 378B
=> [elasticsearch internal] load metadata for docker.io/library/elasticsearch:7.14.0
=> [elasticsearch internal] load .dockerignore
=> -- transferring context: 2B
=> [elasticsearch internal] load build context
=> -- transferring context: 39B
=> [elasticsearch 1/3] FROM docker.io/library/elasticsearch:7.14.0@sha256:81c12e4edd0c5574285679bcb23d47f892ee5e7b74d9b8a79b6f99f1219
CACHED [elasticsearch 2/3] COPY elasticsearch.yml /usr/share/elasticsearch/config/elasticsearch.yml
CACHED [elasticsearch 3/3] RUN chown elasticsearch:elasticsearch /usr/share/elasticsearch/config/elasticsearch.yml
=> [elasticsearch] exporting to image
=> -- exporting layers
=> -- writing image sha256:c950f02f0ba3bd9e5ebdbdff17965dabc72ba56142f4fe470c81974124f0a04f
=> naming to docker.io/library/elasticsearch
=> [setup internal] load build definition from Dockerfile
=> -- transferring Dockerfiles: 22B
=> [backend internal] load metadata for docker.io/library/python:3.9-alpine
=> [backend internal] load build definition from Dockerfile
=> -- transferring Dockerfile: 453B
=> [backend internal] load .dockerignore
=> -- transferring context: 2B
=> [setup internal] load .dockerignore
=> -- transferring context: 2B
=> [setup 1/5] FROM docker.io/library/python:3.9-alpine@sha256:e34551410de80c40494fac7848ab0c796a52f26965c41290a71044fb47fabe
=> [backend internal] load build context
=> -- transferring context: 130B
=> [setup internal] load build context
=> -- transferring context: 71B
=> CACHED [backend 2/5] WORKDIR /app
=> CACHED [setup 3/5] COPY requirements-setup.txt ./requirements.txt
=> CACHED [setup 4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [setup 5/5] COPY setup.py ./setup.py
=> [backend] exporting to image
=> -- exporting layers
=> -- writing image sha256:2c04e931a30e747381d4677f2f7a254ec5da09bde1310a280c055d32b0656
=> naming to docker.io/library/setup
=> CACHED [backend 3/5] COPY requirements-backend.txt ./requirements.txt
=> CACHED [backend 4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [backend 5/5] COPY backend.py ./main.py
=> CACHED [backend 6/5] COPY key.pem .
=> CACHED [backend 7/5] COPY cert.pem .
=> CACHED [backend 8/5] RUN chmod 644 key.pem cert.pem
=> CACHED [backend 9/5] RUN adduser -D appuser
=> [backend] exporting to image
=> -- exporting layers
=> -- writing image sha256:cf0b8990890764cf3179305ba2c124a2df0bf4f07f4cb0b276b074d275ee2ae1d
=> naming to docker.io/library/elasticl1
[+] Running 4/4
✔ Network lab08-app-network
✔ Container lab08-elasticsearch-1 Created
✔ Container lab08-setup-1 Healthy
✔ Container lab08-backend-1 Started
adiborate23@backend-vm:~/lab08$
```

```
adiborate23@backend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
57f6379e9b87   fastapi:2                          "python main.py"        About a minute ago Up About a minute    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp
lab08-backend-1   elasticsearch                      "/bin/tini -- /usr/l..." About a minute ago Up About a minute (healthy)    0.0.0.0:9200->9200/tcp, :::9200->9200/tcp, 0.0.0.0:9300->9300/tcp, :::9300->9300/tcp
lab08-elasticsearch-1
adiborate23@backend-vm:~/lab08$
```

```
adiborate23@backend-vm:~/lab08$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
fastapi              2                  29ef17f2f233f     8 hours ago       73.3MB
setup               latest             2c14e931ae3c       8 hours ago       61.1MB
elasticsearch        latest             c950f8f7c3a3       8 hours ago       1.04GB
adiborate23@backend-vm:~/lab08$
```

- one for the backend (lab08-backend-1) and

- one for Elasticsearch (lab08-elasticsearch-1)

—while three images exist: fastapi:2, elasticsearch, and setup.

The setup image in lab08-setup-1 container was designed to run only once during initialization

- to wait for elasticsearch to be healthy and up
- set up an index ("documents")
- add field mappings (id, text)
- inserts some default documents (first 4 paragraphs from <https://en.wikipedia.org/wiki/India>)
- and then exit.

While its container may no longer be running, the image still remains on the system for reference or potential reuse. So, the container count is less than the image count because not every image must have a continuously running container. The running containers represent active services (backend and Elasticsearch), whereas the images include everything that has been built, including one-time use components like setup.

4. Checking listening ports using `netstat -antp | grep LISTEN`

The screenshot shows a terminal window with the following content:

```

-> [setup] exporting to image
-> => exporting layers
-> => writing image sha256:2c14e931ae3ce747381d4677f2f7a254ec5da09bde131c0a200c055a32b0e5e
-> => naming to docker.io/library/setup
-> CACHED [backend 3/9] COPY requirements-backend.txt ./requirements.txt
-> CACHED [backend 4/9] RUN pip install --no-cache-dir -r requirements.txt
-> CACHED [backend 5/9] COPY backend.py ./main.py
-> CACHED [backend 6/9] COPY key.pem
-> CACHED [backend 7/9] COPY cert.pem
-> CACHED [backend 8/9] RUN chmod 644 key.pem cert.pem
-> CACHED [backend 9/9] RUN adduser -D appuser
-> [backend] exporting to image
-> => exporting layers
-> => writing image sha256:29ef7f2233f17e67e1c38a92f02a2838f4ac3535be55fe6b4c104350b7a0bc
-> => naming to docker.io/library/fastapi:2
[+] Image 2/4
✓ Network lab08_app-network Created
✓ Container lab08-elasticsearch-1 Healthy
✓ Container lab08-backend-1 Started
✓ Container lab08-setup-1 Started
adiborate23@backend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
57f6337e9b87   fastapi:2                          "python main.py"        About a minute ago    Up About a minute    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp
904346c0e80e   elasticsearch                       "/bin/tini -- /usr/l..." About a minute ago    Up About a minute (healthy)    0.0.0.0:9200->9200/tcp, :::9200->9200/tcp, 0.0.0.0:9300->9300/tcp, :::9300
0->9300/tcp    lab08-elasticsearch-1
adiborate23@backend-vm:~/lab08$ docker images
REPOSITORY    TAG        IMAGE ID      CREATED      SIZE
fastapi       2          29ef7f2f233f  8 hours ago  73.3MB
setup         latest    2c14e931ae3c  8 hours ago  61.1MB
elasticsearch latest    c950f02fcb3  8 hours ago  1.04GB
adiborate23@backend-vm:~/lab08$ netstat -antp | grep LISTEN
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:20202        0.0.0.0:*        LISTEN      -
tcp        0      0 0.0.0.0:140467      0.0.0.0:*        LISTEN      -
tcp        0      0 0.0.0.0:9200        0.0.0.0:*        LISTEN      -
tcp        0      0 0.0.0.0:53:53       0.0.0.0:*        LISTEN      -
tcp        0      0 0.0.0.0:9200        0.0.0.0:*        LISTEN      -
tcp        0      0 0.0.0.0:22          0.0.0.0:*        LISTEN      -
tcp        0      0 0.0.0.0:9567        0.0.0.0:*        LISTEN      -
tcp6       0      0 :::20201            :::*              LISTEN      -
tcp6       0      0 :::9200             :::*              LISTEN      -
tcp6       0      0 :::9300             :::*              LISTEN      -
tcp6       0      0 :::22               :::*              LISTEN      -
tcp6       0      0 :::9567             :::*              LISTEN      -
adiborate23@backend-vm:~/lab08$

```

5. `docker inspect` on backend container

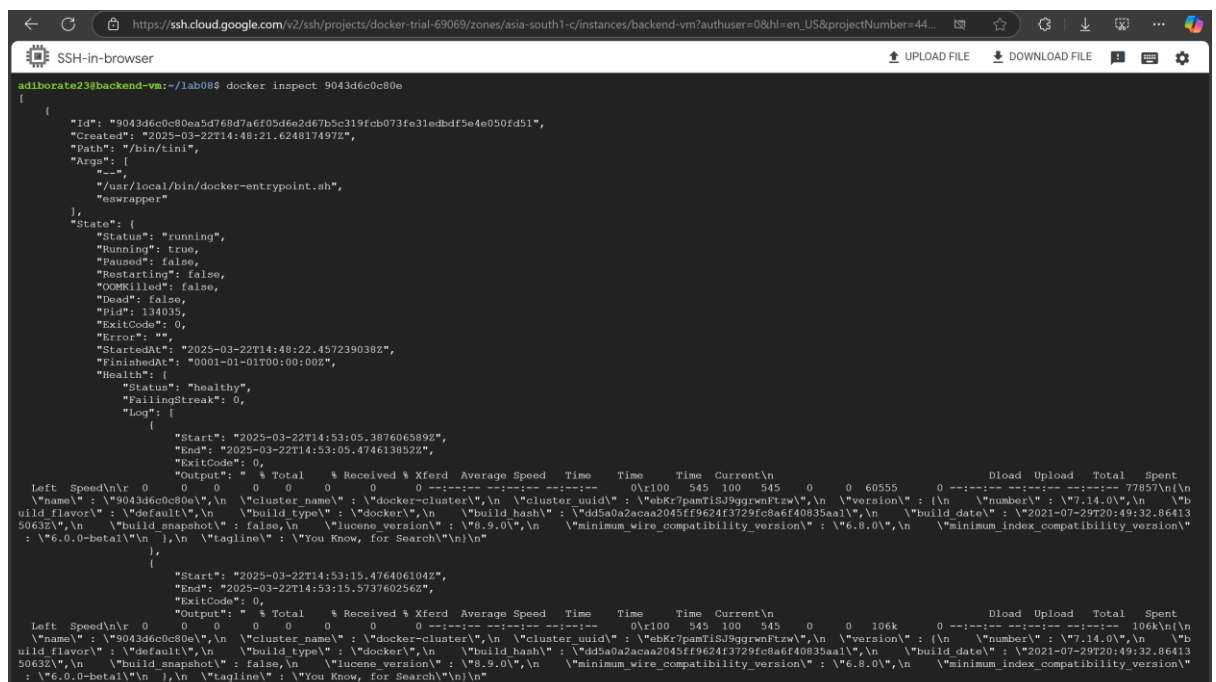
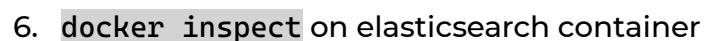
```
SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...
UPLOAD FILE DOWNLOAD FILE

adiborate23@backend-vm:~/lab08$ docker inspect 57f6337e9b87
[
  {
    "Id": "57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a",
    "Created": "2025-03-22T14:48:21.687277984Z",
    "Path": "python",
    "Args": [
      "main.py"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 134478,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2025-03-22T14:48:55.222993751Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:29ef7f2f233f17e67e1c38a92f02e2838f4ac3535be55fe6ba4c194350b7a0bc",
    "ResolvConfPath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/hostname",
    "HostsPath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/hosts",
    "LogPath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a-jso
n.log",
    "Name": "/lab08-backend-1",
    "RestartCount": 0,
    "Driver": "overlay2",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "docker-default",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
        "Config": {}
      },
      "NetworkMode": "lab08_app-network",
      "PortBindings": {
        "9567/tcp": [
          {
            "HostIp": "",
            "HostPort": "9567"
          }
        ]
      },
      "RestartPolicy": {
        "Name": "no",
        "MaximumRetryCount": 0
      },
      "AutoRemove": false,
      "VolumeDriver": "",
      "VolumesFrom": null,
      "ConsoleSize": {
        0,
        0
      },
      "CapAdd": null,
      "CapDrop": null,
      "CgroupnsMode": "host",
      "Dns": null,
      "DnsOptions": null,
      "DnsSearch": null,
      "ExtraHosts": {},
      "GroupAdd": null,
      "IpcMode": "private",
      "Cgroup": "",
      "Links": null,
      "OomScoreAdj": 0,
      "PidMode": "",
      "Privileged": false,
      "PublishAllPorts": false,
      "ReadonlyRootfs": false,
      "SecurityOpt": null,
      "UTSMode": "",
      "UsernsMode": "",
      "ShmSize": 67108864,
      "Runtime": "runc",
      "Isolation": "",
      "CpuShares": 0,
      "Memory": 0,
      "NanoCpus": 0,
      "CgroupParent": "",
      "BlkioWeight": 0,
      "BlkioWeightDevice": null,
      "BlkioDeviceReadBps": null,
      "BlkioDeviceWriteBps": null,
      "BlkioDeviceReadOps": null,
      "BlkioDeviceWriteOps": null,
      "CpuPeriod": 0,

```

```
SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...
UPLOAD FILE DOWNLOAD FILE

"CPUQuota": 0,
"CPURealtimePeriod": 0,
"CPURealtimeRuntime": 0,
"CpusetCpus": "",
"CpusetMems": "",
"Devices": null,
"DeviceGroupRules": null,
"DeviceRequests": null,
"MemoryReservation": 0,
"MemorySwap": 0,
"MemorySwappiness": null,
"OomKillDisable": false,
"PidsLimit": null,
"Ulimits": null,
"CPUCount": 0,
"CPUPercent": 0,
"IOMaximumIops": 0,
"IOMaximumBandwidth": 0,
"MaskedPaths": [
  "/proc/asound",
  "/proc/acpi",
  "/proc/kcore",
  "/proc/kmsg",
  "/proc/latency_stats",
  "/proc/timer_list",
  "/proc/timer_stats",
  "/proc/sched_debug",
  "/proc/pci",
  "/sys/firmware",
  "/sys/devices/virtual/powercap"
],
"ReadOnlyPaths": [
  "/proc/bus",
  "/proc/fs",
  "/proc/iq",
  "/proc/sys",
  "/proc/sysrq-trigger"
],
"GraphDriver": {
  "Data": {
    "LowerDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4-init/diff:/var/lib/docker/overlay2/quh1qoo56kh9908e0eazq85d/diff:/var/lib/docker/overlay2/yrordamfz0znflualb8cgsz/diff:/var/lib/docker/overlay2/r1051kq3i73597bftb91xflpn/diff:/var/lib/docker/overlay2/d0xtuxlubi6f92d5tg5zyyt7/diff:/var/lib/docker/overlay2/ipyagf9wvba2bus0yyqlhoh/diff:/var/lib/docker/overlay2/oohilfcjdjdx4vkjx9oug6gk8/diff:/var/lib/docker/overlay2/naelt2k4qzymjzbfha8x1hhq1/diff:/var/lib/docker/overlay2/koiyy4xyimbhnmzj1a4gaeu2r/diff:/var/lib/docker/overlay2/2d6231c527d9836cc30741657255d5113ee27a263a8aed40997d591ea4cd9b8a/diff:/var/lib/docker/overlay2/77915050d12592ce33858de478a501191870f3c28a2e440b499b1d8/diff:/var/lib/docker/overlay2/ae266fc7df2c67edb0db7bd71709e71f4d82a62cf104e2c46dc9ac83ac093e3/diff:/var/lib/docker/overlay2/a2d74b6295c79e78a3288943f3ade6532bba84202d7a334a544b4859423403d/diff",
    "MergedDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4/merged",
  }
},
"UpperDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4/diff",
"WorkDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4/work"
},
"Name": "overlay2"
},
"Mounts": [],
"Config": {
  "Hostname": "57f6337e9b07",
  "Domainname": "",
  "User": "appuser",
  "AttachStdin": false,
  "AttachStdout": true,
  "AttachStderr": true,
  "ExposedPorts": {
    "9567/tcp": {}
  },
  "Tty": false,
  "OpenStdin": false,
  "StdinOnce": false,
  "Env": [
    "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
    "LANG=C.UTF-8",
    "GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
    "PYTHON_VERSION=3.9.21",
    "PYTHON_SHA256=312ef59592c9b0d798584755f2bf7b081falca35ce7a6fea980108d752a05bb1"
  ],
  "Cmd": [
    "python",
    "main.py"
  ],
  "Image": "fantapi2",
  "Volumes": null,
  "WorkingDir": "/app",
  "Entrypoint": null,
  "OnBuild": null,
  "Labels": {
    "com.docker.compose.config-hash": "47a8207a3c207a206cb2746a0c17ac7487369398bd5fddede50f982637fe70",
    "com.docker.compose.container-number": "1",
    "com.docker.compose.depends_on": "elasticsearch:service_healthy:false",
    "com.docker.compose.image": "sha256:29ef7f2f233f17e7e1c38a92f02e2838f4ac3535be55fe6ba4c184350b7a0bc",
    "com.docker.compose.oneoff": "false",
    "com.docker.compose.project": "lab08",
    "com.docker.compose.project.config_files": "/home/adiborate23/lab08/docker-compose.yml",
    "com.docker.compose.project.working_dir": "/home/adiborate23/lab08",
    "com.docker.compose.service": "backend",
    "com.docker.compose.version": "2.20.2"
  }
}
},
```



[illegible]


```
SSH-in-browser

"CapDrop": null,
"CgroupnsMode": "host",
"Dns": null,
"DnsOptions": null,
"DnsSearch": null,
"ExtraHosts": {},
"GroupAdd": null,
"IpcMode": "private",
"Cgroup": "",
"Links": null,
"OomScoreAdj": 0,
"PidMode": "",
"Privileged": false,
"PublishAllPorts": false,
"ReadOnlyRootfs": false,
"SecurityOpt": null,
"UTSMode": "",
"UsernsMode": "",
"ShmSize": 67108864,
"Runtime": "runc",
"Isolation": "",
"CpuShares": 0,
"Memory": 0,
"NanoCpus": 0,
"CgroupParent": "",
"BlkioWeight": 0,
"BlkioWeightDevice": null,
"BlkioDeviceReadBps": null,
"BlkioDeviceWriteBps": null,
"BlkioDeviceReadIops": null,
"BlkioDeviceWriteIops": null,
"CpuPeriod": 0,
"CpuQuota": 0,
"CpuRealtimePeriod": 0,
"CpuRealtimeRuntime": 0,
"CpusetCpus": "",
"CpusetMems": "",
"Devices": null,
"DeviceCgroupRules": null,
"DeviceRequests": null,
"MemoryReservation": 0,
"MemorySwap": 0,
"MemorySwappiness": null,
"OomKillDisable": false,
"PidsLimit": null,
"Ulimits": null,
"CpuCount": 0,
"CpuPercent": 0,

"IOMaximumOps": 0,
"IOMaximumBandwidth": 0,
"Mounts": [
  {
    "Type": "volume",
    "Source": "lab08_esdata",
    "Target": "/usr/share/elasticsearch/data",
    "VolumeOptions": {}
  }
],
"MaskedPaths": [
  "/proc/anousd",
  "/proc/aspi",
  "/proc/kcore",
  "/proc/keys",
  "/proc/latency_stats",
  "/proc/timer_list",
  "/proc/timer_stats",
  "/proc/sched debug",
  "/proc/scsi",
  "/sys/firmware",
  "/sys/devices/virtual/powercap"
],
"ReadOnlyPaths": [
  "/proc/bus",
  "/proc/fs",
  "/proc/irq",
  "/proc/sys",
  "/proc/sysrq-trigger"
],
"GraphDriver": {
  "Data": {
    "LowerDir": [
      "/var/lib/docker/overlay2/b1fe22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5-init/diff:/var/lib/docker/overlay2/7xhjhwqe3k2z46xa14tbl47gt/diff:/var/lib/docker/overlay2/001135evpx6lpm3x2yp7qqr/diff:/var/lib/docker/overlay2/55c3f3982b54437440049096c9643b7cbf8ec1a08f69a3b6208920bd9a1bf/diff:/var/lib/docker/overlay2/68f040264c5b7f431c0e70977f2cd09c7ca0452017ae63702f18474aa6877aa4e2/diff:/var/lib/docker/overlay2/122f6800225423cc2e86a85bc6e5502a8855ba0079e60c658b75ad5f5fc2f8/diff:/var/lib/docker/overlay2/fcc1445be07038e884b7bba6f96f6bd5608a9954936ae53cb52981ae504f096/diff:/var/lib/docker/overlay2/f093889aa53e09baff0e1eb99f89b718e2371adcccc133a8f6c0a61ae2691a40/diff:/var/lib/docker/overlay2/4b366a8a0441b9a1455925234d7c61f2ca5f9819396dede1cd1d2521b8a13b36/diff:/var/lib/docker/overlay2/36c09f9274384e0e2b4fe9d2d50b241da0910ac254c46fcff50fc5d2cde689a6/diff",
      "MergedDir": "/var/lib/docker/overlay2/b1fe22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5/merged",
      "UpperDir": "/var/lib/docker/overlay2/b1fe22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5/diff",
      "WorkDir": "/var/lib/docker/overlay2/b1fe22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5/work"
    ],
    "Name": "overlay2"
  },
  "Mounts": [
    {
      "Type": "volume",
```

```
SSH-in-browser

"Name": "lab08_esdata",
"Source": "/var/lib/docker/volumes/lab08_esdata_data",
"Destination": "/usr/share/elasticsearch/data",
"Driver": "local",
"Mode": "z",
"RW": true,
"Propagation": ""
},
"Config": {
  "Hostname": "9043d6c0c80e",
  "Domainname": "",
  "User": "elasticsearch",
  "AttachStdin": false,
  "AttachStdout": true,
  "AttachStderr": true,
  "ExposedPorts": {
    "9200/tcp": {},
    "9300/tcp": {}
  },
  "Tty": false,
  "OpenStdin": false,
  "StdinOnce": false,
  "Env": [
    "discovery.type=single-node",
    "ES_JAVA_OPTS=-Xms512m -Xmx512m",
    "PATH=/usr/share/elasticsearch/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
    "ELASTIC_CONTAINER=true"
  ],
  "Cmd": [
    "eswrapper"
  ],
  "Healthcheck": {
    "Test": [
      "CMD",
      "curl",
      "-f",
      "http://localhost:9200"
    ],
    "Interval": 1000000000,
    "Timeout": 5000000000,
    "StartPeriod": 20000000000,
    "Retries": 5
  },
  "Image": "elasticsearch",
  "Volumes": null,
  "WorkingDir": "/usr/share/elasticsearch",
  "Entrypoint": [
    "/bin/tini",
    "-s",
    "/usr/local/bin/docker-entrypoint.sh"
  ],
  "OnBuild": null,
  "Labels": {
    "com.docker.compose.config-hash": "d37824d669286c91a4c612e2264ba054289988f663217bd17e96cc0ca3194c59",
    "com.docker.compose.container-number": "1",
    "com.docker.compose.depends_on": "",
    "com.docker.compose.image": "sha256:c950f82fcb3ed965a8b0d0ff7796fdebc72b956142f4fe470c81974124f0a86f",
    "com.docker.compose.offset": "false",
    "com.docker.compose.project": "lab08",
    "com.docker.compose.project.config_files": "/home/adiborate23/lab08/docker-compose.yml",
    "com.docker.compose.project.working_dir": "/home/adiborate23/lab08",
    "com.docker.compose.service": "elasticsearch",
    "com.docker.compose.version": "2.20.2",
    "org.label-schema.build-date": "2021-07-29T20:49:32.864135063Z",
    "org.label-schema.license": "Elastic-License-2.0",
    "org.label-schema.name": "Elasticsearch",
    "org.label-schema.schema-version": "7.10",
    "org.label-schema.url": "https://www.elastic.co/products/elasticsearch",
    "org.label-schema.usage": "https://www.elastic.co/guide/en/elasticsearch/reference/index.html",
    "org.label-schema.vcs-ref": "dd5a0a2acaa2045ff9624f3729fcb8a6f40835aal",
    "org.label-schema.vcs-uri": "https://github.com/elastic/elasticsearch",
    "org.label-schema.vendor": "Elastic",
    "org.label-schema.version": "7.14.0",
    "org.opencontainers.image.created": "2021-07-29T20:49:32.864135063Z",
    "org.opencontainers.image.documentation": "https://www.elastic.co/guide/en/elasticsearch/reference/index.html",
    "org.opencontainers.image.license": "Elastic-License-2.0",
    "org.opencontainers.image.revision": "dd5a0a2acaa2045ff9624f3729fcb8a6f40835aal",
    "org.opencontainers.image.source": "https://github.com/elastic/elasticsearch",
    "org.opencontainers.image.title": "Elasticsearch",
    "org.opencontainers.image.url": "https://www.elastic.co/products/elasticsearch",
    "org.opencontainers.image.vendor": "Elastic",
    "org.opencontainers.image.version": "7.14.0"
  }
},
"NetworkSettings": {
  "Bridge": "",
  "SandboxID": "4f692be57d90aa1762226e08f57aab484f3e113b9af5b4d6b2014a6ec9aeb29",
  "SandboxKey": "/var/run/docker/netns/4f692be57d90",
  "Ports": {
    "9200/tcp": [
      {
        "HostIp": "0.0.0.0",
        "HostPort": "9200"
      }
    ]
  }
}
```

```

    "HostIp": "::",
    "HostPort": "9200"
  },
  "9300/tcp": [
    {
      "HostIp": "0.0.0.0",
      "HostPort": "9300"
    },
    {
      "HostIp": "::",
      "HostPort": "9300"
    }
  ],
  "HairpinMode": false,
  "LinkLocalIPv6Address": "",
  "LinkLocalIPv6PrefixLen": 0,
  "SecondaryIPAddresses": null,
  "SecondaryIPv6Addresses": null,
  "EndpointID": "",
  "Gateway": "",
  "GlobalIPv6Address": "",
  "GlobalIPv6PrefixLen": 0,
  "IPAddress": "",
  "IPPrefixLen": 0,
  "IPv6Gateway": "",
  "MacAddress": "",
  "Networks": {
    "lab08_app-network": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": [
        "lab08-elasticsearch-1",
        "elasticsearch"
      ],
      "MacAddress": "02:42:ac:18:00:02",
      "NetworkID": "2009070cc691fc2dd5b80352ec8fa9505b210ee04c0607312baefcc2285e6b74",
      "EndpointID": "6f09fc94ff7b522d248bf1d9c7db8b4f56d673083a1910c029991283c395df0a",
      "Gateway": "172.24.0.1",
      "IPAddress": "172.24.0.2",
      "IPPrefixLen": 16,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "DriverOpts": null,
      "DNSNames": [
        "lab08-elasticsearch-1",
        "elasticsearch",
        "9043d6cb080e"
      ]
    }
  ]
}
}
}
}
}
adiborate23@backend-vm:~/lab08$

```

✓ Running FastAPI and Elasticsearch on Specific Ports

1. FastAPI (Frontend and Backend)

Both FastAPI services (frontend and backend) are explicitly configured to run on port **9567**

- In **main.py** (Frontend):

```

if __name__ == "__main__":
    uvicorn.run(
        app,
        host="0.0.0.0",
        port=9567,
        ssl_keyfile="key.pem",
        ssl_certfile="cert.pem"
    )

```

- In **backend.py** (Backend):

```

# Run the FastAPI app
if __name__ == "__main__":
    uvicorn.run(app, host="0.0.0.0", port=9567)

```

- Dockerfile exposes the port:

```
EXPOSE 9567
```

- Docker Compose maps the container port to host:

```
ports:
  - "9567:9567"
```

2. Elasticsearch (Private Access Only)

Elasticsearch runs on standard ports internally: 9200 (for REST API) and 9300 (for node communication), but they are **not published** to the host machine. This is done to :

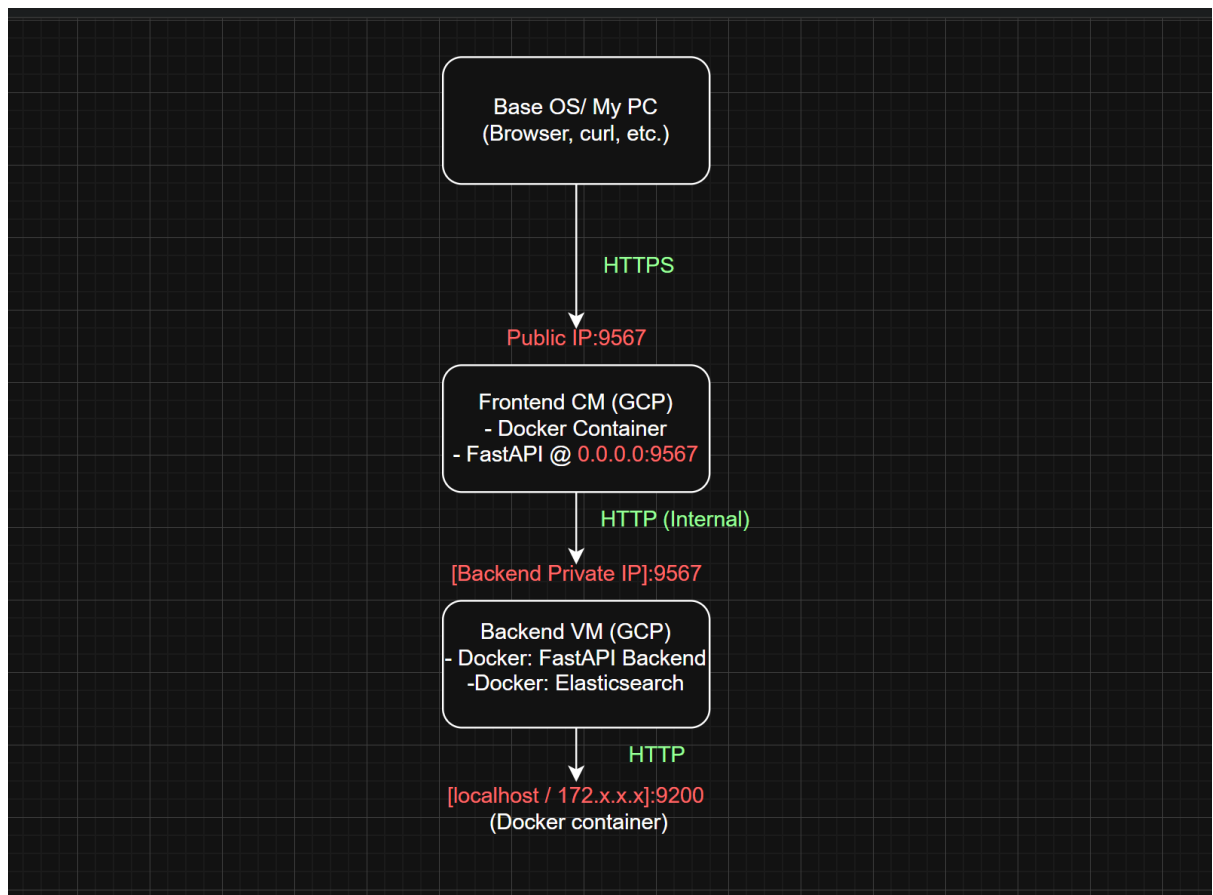
- Prevent external or frontend access (even via private IP)
- Limit Elasticsearch access to the backend container only
- Improve security by reducing attack surface

Docker Compose (Backend VM):

```
# Removed public port bindings to restrict access
# ports:
#   - "9200:9200"
#   - "9300:9300"
```

Elasticsearch still works fine because the backend and Elasticsearch containers are on the same internal Docker network (`app-network`), allowing internal communication.

✓ Check the accessibility of ports



1. Inside frontend-vm:

a. curl <http://10.160.0.8:9567/docs>

```
adiborate23@frontend-vm:~$ curl http://10.160.0.8:9567/docs
<!DOCTYPE html>
<html>
<head>
<link type="text/css" rel="stylesheet" href="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css">
<link rel="shortcut icon" href="https://fastapi.tiangolo.com/img/favicon.png">
<title>FastAPI - Swagger UI</title>
</head>
<body>
<div id="swagger-ui">
</div>
<script src="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js"></script>
<!-- 'SwaggerUIBundle' is now available on the page -->
<script>
const ui = SwaggerUIBundle({
  url: '/openapi.json',
  'dom_id': '#swagger-ui',
  'layout': 'BaseLayout',
  'deepLinking': true,
  'showExtensions': true,
  'showCommonExtensions': true,
  oauth2RedirectUrl: window.location.origin + '/docs/oauth2-redirect',
  presets: [
    SwaggerUIBundle.presets.apis,
    SwaggerUIBundle.SwaggerUIStandalonePreset
  ],
})
</script>
</body>
</html>
adiborate23@frontend-vm:~$
```

--WORKED: Shows frontend can reach backend via internal IP. This confirms inter-VM internal networking is correctly set up.

b. curl -k <https://0.0.0.0:9567/>

```
adiborate23@frontend-vm:~/lab08/frontend$ curl -k https://0.0.0.0:9567/
<!DOCTYPE html>
<html>
<head>
<title>Elasticsearch Interface</title>
</head>
<body>
<h1>Elasticsearch Document Manager</h1>
<h2>Search Document</h2>
<form action="/get" method="post">
  <input type="text" name="query" placeholder="Enter search query" required>
  <button type="submit">Get</button>
</form>
<h2>Insert Document</h2>
<form action="/insert" method="post">
  <textarea name="content" rows="5" cols="50" placeholder="Enter document content" required></textarea><br>
  <button type="submit">Insert</button>
</form>
<div id="result" style="margin-top: 20px;">
  <h2>Result:</h2>
  <p>No results to display.</p>
</div>
</body>
</html>
adiborate23@frontend-vm:~/lab08/frontend$
```

--WORKED: Confirms FastAPI in frontend is bound to all interfaces.

c. curl -k -X POST <https://34.47.186.152:9567/insert> \
--data-urlencode "content=test from frontend"

```
adiborate23@frontend-vm:~/lab08$ curl -k -X POST http://34.47.186.152:9567/insert \
> --data-urlencode "content=test from frontend"
<!DOCTYPE html>
<html>
<head>
<title>Elasticsearch Interface</title>
</head>
<body>
<h1>Elasticsearch Document Manager</h1>

<h2>Search Document</h2>
<form action="/get" method="post">
<input type="text" name="query" placeholder="Enter search query" required>
<button type="submit">Get</button>
</form>

<h2>Insert Document</h2>
<form action="/insert" method="post">
<textarea name="content" rows="5" cols="50" placeholder="Enter document content" required></textarea><br>
<button type="submit">Insert</button>
</form>

<div id="result" style="margin-top: 20px;">
<h2>Result:</h2>

<p>Document inserted successfully with ID: 9tp4v5UBylE19jowNAd0</p>

</div>
</body>
</html>adiborate23@frontend-vm:~/lab08$
```

--WORKED: Confirms public access to frontend is functional.

d. `curl -v --connect-timeout 5 http://34.47.206.180:9567`

```
adiborate23@frontend-vm:~/lab08/frontend$ curl -v --connect-timeout 5 http://34.47.206.180:9567
* Trying 34.47.206.180:9567...
* TCP_NODELAY set
* Connection timed out after 5000 milliseconds
* Closing connection 0
curl: (28) Connection timed out after 5000 milliseconds
```

-- FAILED: Confirms public access to backend FastAPI is blocked, which is expected.

e. `curl -v --connect-timeout 5 http://34.47.206.180:9200`

and

`curl -v --connect-timeout 5 http://34.47.206.180:9300`

```
adiborate23@frontend-vm:~/lab08/frontend$ curl -v --connect-timeout 5 http://34.47.206.180:9200
* Trying 34.47.206.180:9200...
* TCP_NODELAY set
* Connection timed out after 5000 milliseconds
* Closing connection 0
curl: (28) Connection timed out after 5000 milliseconds
adiborate23@frontend-vm:~/lab08/frontend$

adiborate23@frontend-vm:~/lab08$ curl -v --connect-timeout 5 http://34.47.206.180:9300
* Trying 34.47.206.180:9300...
* TCP_NODELAY set
* Connection timed out after 5001 milliseconds
* Closing connection 0
curl: (28) Connection timed out after 5001 milliseconds
```

-- FAILED: Confirms public access to Elasticsearch is blocked — again, expected.

f. `curl http://10.160.0.8:9200`

and

`curl http://10.160.0.8:9300`

```
adiborate23@frontend-vm:~/lab08$ curl http://10.160.0.8:9200
curl: (7) Failed to connect to 10.160.0.8 port 9200: Connection refused
adiborate23@frontend-vm:~/lab08$ curl http://10.160.0.8:9300
curl: (7) Failed to connect to 10.160.0.8 port 9300: Connection refused
adiborate23@frontend-vm:~/lab08$
```

-- FAILED: Confirms Elasticsearch cannot be accessed by frontend via private IP too— again, expected because frontend should only communicate with backend (that too only via private IP) and not directly with elasticsearch.

2. Inside backend-vm:

a. `curl https://localhost:9200`

```
adiborate23@backend-vm:~/lab08$ curl http://localhost:9200
{
  "name" : "9043d6c0c80e",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "ebKr/pamTjS39ggrwnFtzv",
  "version" : {
    "number" : "7.14.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1",
    "build_date" : "2021-07-29T20:49:32.864135063Z",
    "build_snapshot" : false,
    "lucene_version" : "8.9.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
```

-- WORKED: Confirms Elasticsearch is running and accessible locally from the backend container. Backend can communicate with Elasticsearch.

b. `curl -k https://10.160.0.7:9567/docs`

```
adiborate23@backend-vm:~$ curl -k https://10.160.0.7:9567/docs
<DOCTYPE html>
<html>
<head>
<link type="text/css" rel="stylesheet" href="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css">
<link rel="shortcut icon" href="https://fastapi.tiangolo.com/img/favicon.png">
<title>FastAPI - Swagger UI</title>
</head>
<body>
<div id="swagger-ui">
</div>
<script src="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js"></script>
<!-- 'SwaggerUIBundle' is now available on the page -->
<script>
const ui = SwaggerUIBundle({
  url: '/openapi.json',
  'dom_id': '#swagger-ui',
  'layout': 'BaseLayout',
  'deepLinking': true,
  'showExtensions': true,
  'showCommonExtensions': true,
  oauth2RedirectUrl: window.location.origin + '/docs/oauth2-redirect',
  presets: [
    SwaggerUIBundle.presets.apis,
    SwaggerUIBundle.SwaggerUIStandalonePreset
  ],
})
</script>
</body>
</html>
adiborate23@backend-vm:~$
```

-- WORKED: Backend can reach frontend internally via private IP.

✓ Website Working Demo

1. Searching in existing document (get query)

Elasticsearch Document Manager

Search Document

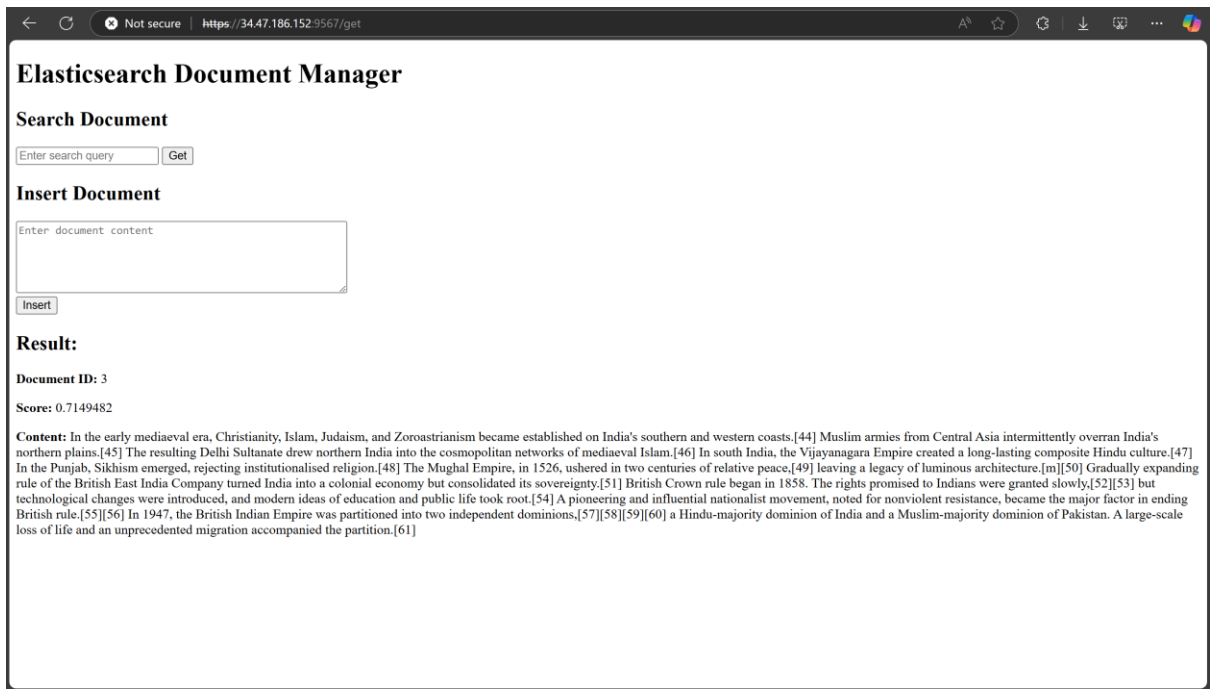
India

Insert Document

Enter document content

Result:

No results to display.



The screenshot shows a web browser window with the URL `https://34.47.186.152:9567/get`. The page title is "Elasticsearch Document Manager". It has two main sections: "Search Document" and "Insert Document". The "Search Document" section has a text input field with "Enter search query" and a "Get" button. The "Insert Document" section has a text area with "Enter document content" and an "Insert" button. Below these is a "Result:" section. It displays "Document ID: 3" and "Score: 0.7149482". The "Content:" field shows a long paragraph of text about the history of India, mentioning the Delhi Sultanate, the Vijayanagara Empire, the Mughal Empire, and the British East India Company.

Elasticsearch Document Manager

Search Document

Enter search query

Insert Document

Enter document content

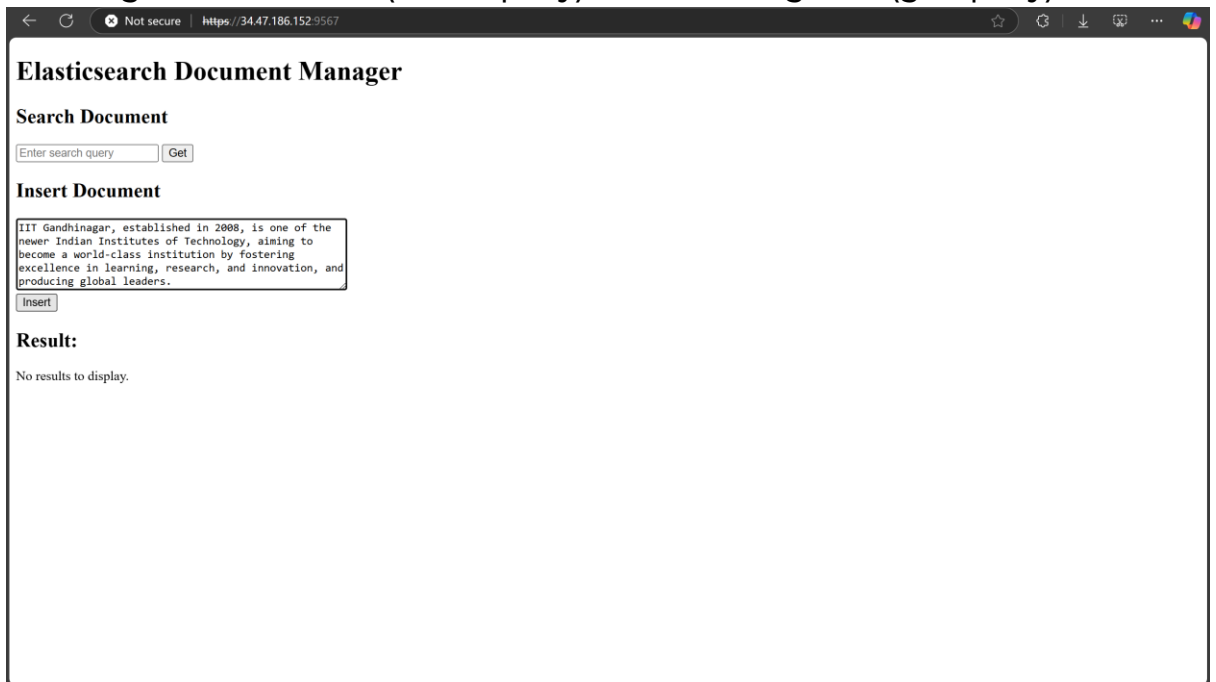
Result:

Document ID: 3

Score: 0.7149482

Content: In the early mediaeval era, Christianity, Islam, Judaism, and Zoroastrianism became established on India's southern and western coasts.[44] Muslim armies from Central Asia intermittently overran India's northern plains.[45] The resulting Delhi Sultanate drew northern India into the cosmopolitan networks of mediaeval Islam.[46] In south India, the Vijayanagara Empire created a long-lasting composite Hindu culture.[47] In the Punjab, Sikhism emerged, rejecting institutionalised religion.[48] The Mughal Empire, in 1526, ushered in two centuries of relative peace.[49] leaving a legacy of luminous architecture.[m][50] Gradually expanding rule of the British East India Company turned India into a colonial economy but consolidated its sovereignty.[51] British Crown rule began in 1858. The rights promised to Indians were granted slowly,[52][53] but technological changes were introduced, and modern ideas of education and public life took root.[54] A pioneering and influential nationalist movement, noted for nonviolent resistance, became the major factor in ending British rule.[55][56] In 1947, the British Indian Empire was partitioned into two independent dominions.[57][58][59][60] a Hindu-majority dominion of India and a Muslim-majority dominion of Pakistan. A large-scale loss of life and an unprecedented migration accompanied the partition.[61]

2. Inserting new document (insert query) and searching for it (get query)



The screenshot shows the same web browser window as the previous one. The "Insert Document" section now contains a text area with the text: "IIT Gandhinagar, established in 2008, is one of the newer Indian Institutes of Technology, aiming to become a world-class institution by fostering excellence in learning, research, and innovation, and producing global leaders." The "Insert" button is visible. The "Result:" section now displays "No results to display."

Elasticsearch Document Manager

Search Document

Enter search query

Insert Document

IIT Gandhinagar, established in 2008, is one of the newer Indian Institutes of Technology, aiming to become a world-class institution by fostering excellence in learning, research, and innovation, and producing global leaders.

Result:

No results to display.

← ↻ Not secure | https://34.47.186.152:9567/insert

Elasticsearch Document Manager

Search Document

Insert Document

Result:

Document inserted successfully with ID: ACL6vpUBXeqfo8OnpaSY

← ↻ Not secure | https://34.47.186.152:9567/insert

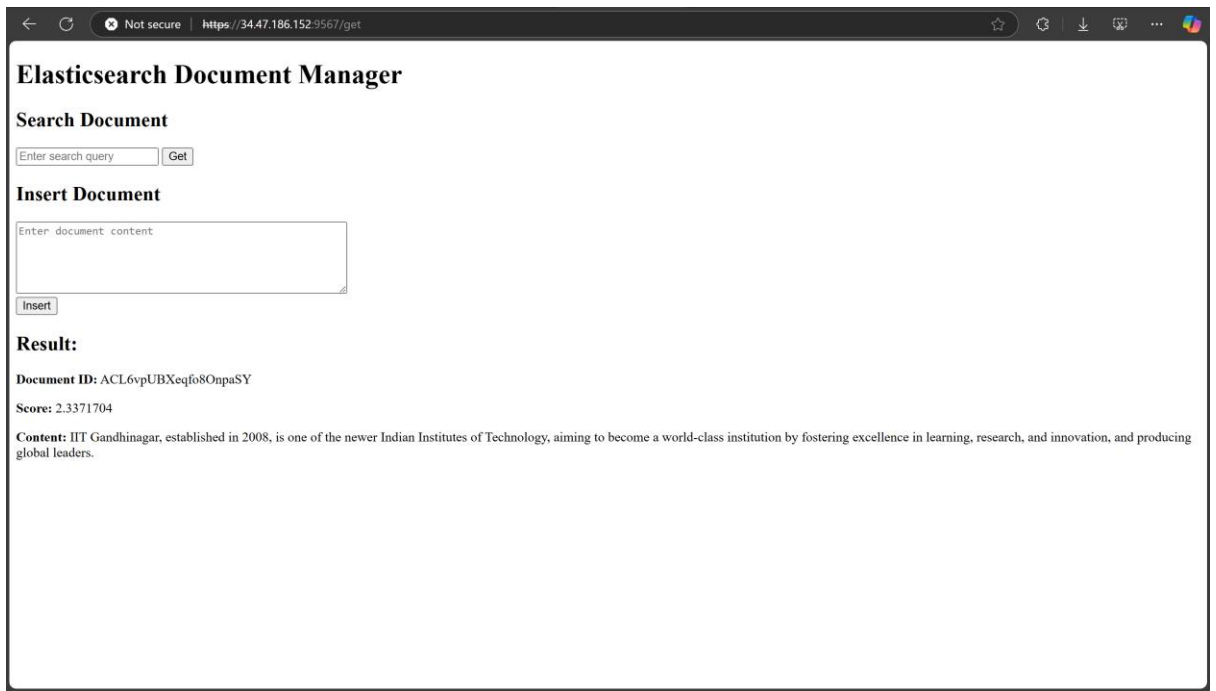
Elasticsearch Document Manager

Search Document

Insert Document

Result:

Document inserted successfully with ID: ACL6vpUBXeqfo8OnpaSY



The screenshot shows a web browser window with the address bar displaying "https://34.47.186.152:9567/get". The page title is "Elasticsearch Document Manager". It features two main sections: "Search Document" and "Insert Document". The "Search Document" section has a text input field containing "Enter search query" and a "Get" button. The "Insert Document" section has a larger text input field containing "Enter document content" and an "Insert" button. Below these sections, the "Result:" section displays the following information: "Document ID: ACL6vpUBXeqfo8OnpaSY", "Score: 2.3371704", and "Content: IIT Gandhinagar, established in 2008, is one of the newer Indian Institutes of Technology, aiming to become a world-class institution by fostering excellence in learning, research, and innovation, and producing global leaders."

Elasticsearch Document Manager

Search Document

Enter search query

Insert Document

Enter document content

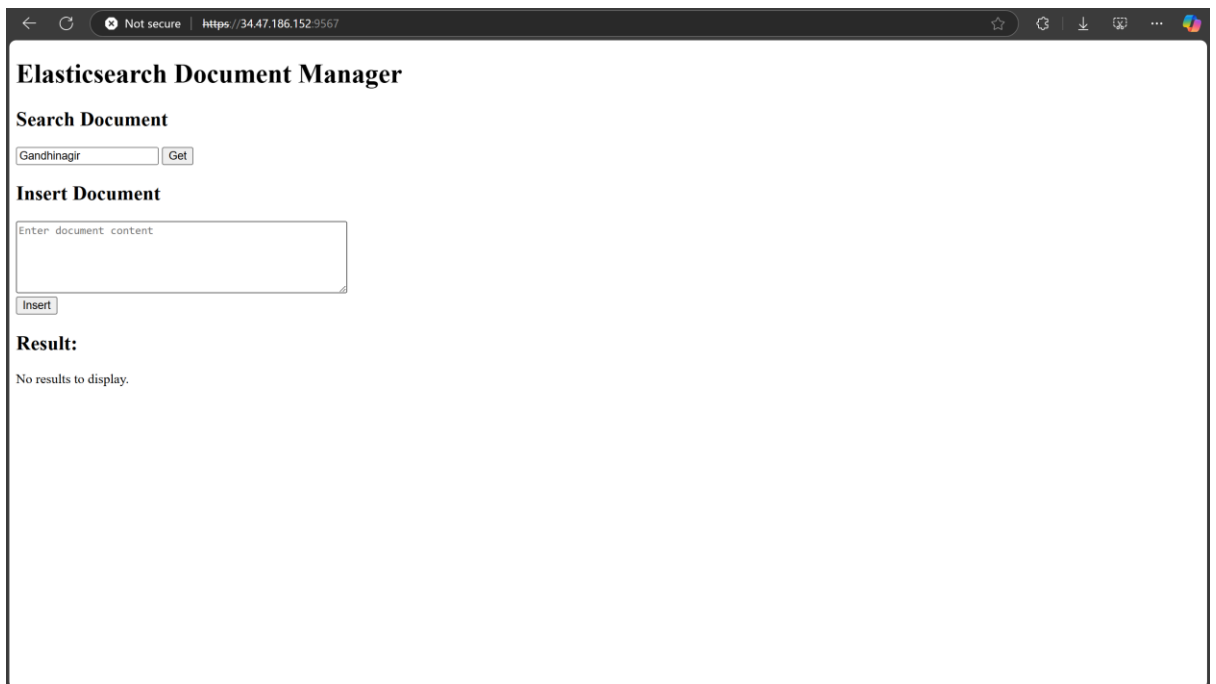
Result:

Document ID: ACL6vpUBXeqfo8OnpaSY

Score: 2.3371704

Content: IIT Gandhinagar, established in 2008, is one of the newer Indian Institutes of Technology, aiming to become a world-class institution by fostering excellence in learning, research, and innovation, and producing global leaders.

3. Checking the ability of elasticsearch to handle typos in queries



The screenshot shows the same web browser window as the previous one, but the "Search Document" section now has a text input field containing "Gandhinagir" (a typo for Gandhinagar) and the "Get" button. The "Result:" section now displays "No results to display."

Elasticsearch Document Manager

Search Document

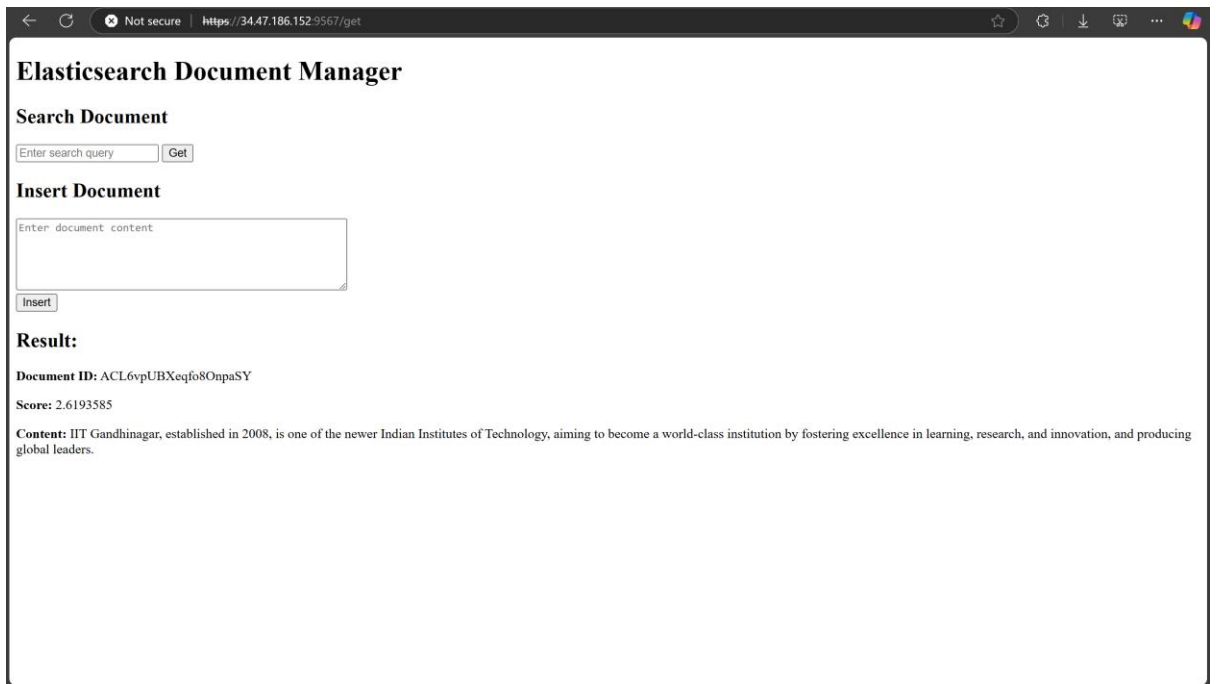
Gandhinagir

Insert Document

Enter document content

Result:

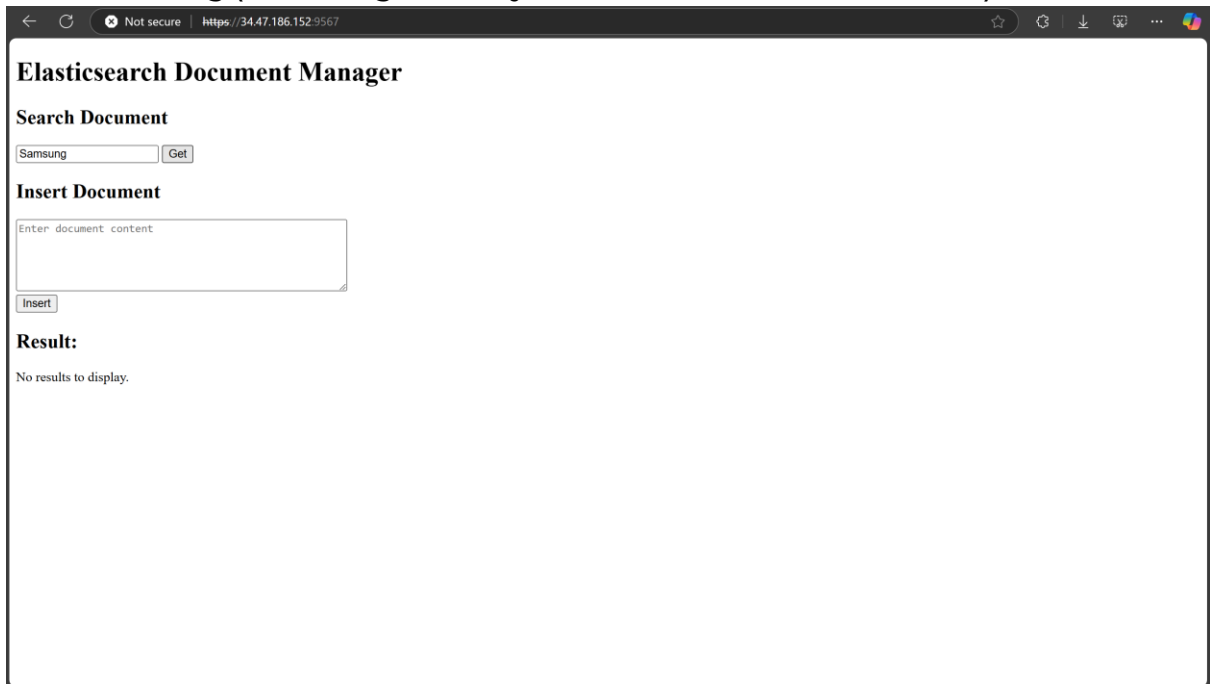
No results to display.



The screenshot shows a web browser window with the address bar displaying "https://34.47.186.152:9567/get". The page title is "Elasticsearch Document Manager". Under the "Search Document" section, the "Enter search query" field contains "Samsung" and the "Get" button is visible. The "Insert Document" section has an empty "Enter document content" text area and an "Insert" button. The "Result:" section displays the following information:

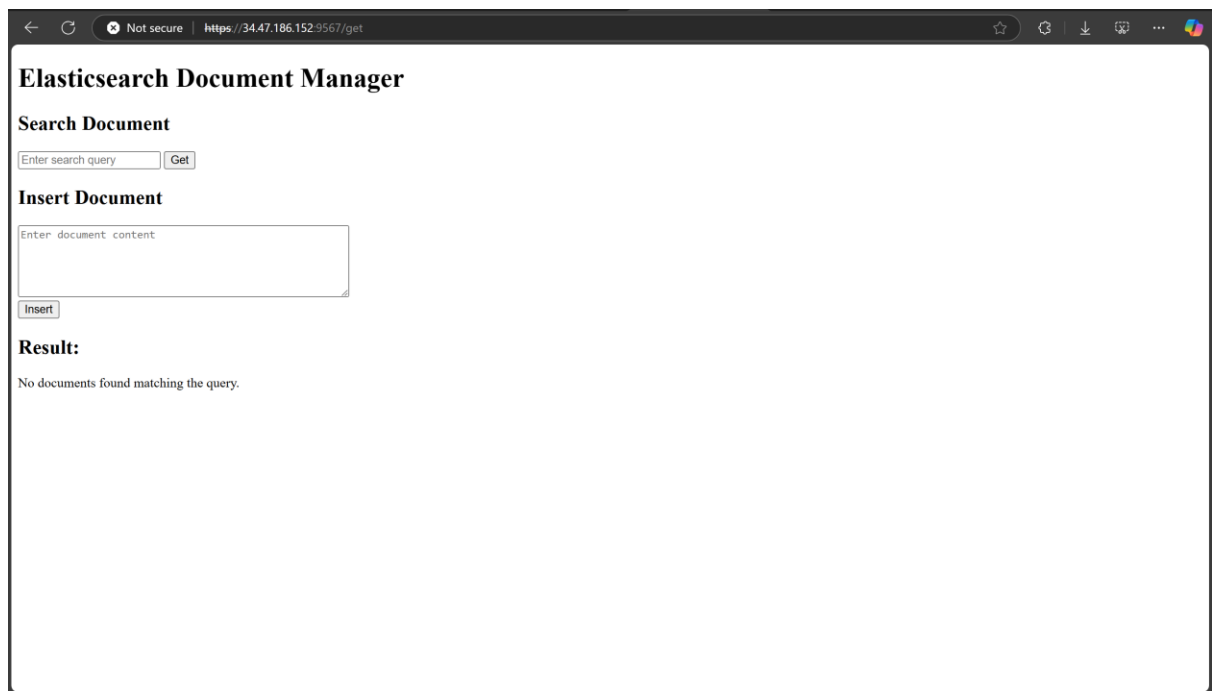
Document ID: ACL6vpUBXeqfo8OnpaSY
Score: 2.6193585
Content: IIT Gandhinagar, established in 2008, is one of the newer Indian Institutes of Technology, aiming to become a world-class institution by fostering excellence in learning, research, and innovation, and producing global leaders.

4. Error Handling (Searching for a key that didn't exist in documents)



The screenshot shows the same web browser window as the previous one. In the "Search Document" section, the "Enter search query" field now contains "Samsung" and the "Get" button is visible. The "Insert Document" section remains the same. The "Result:" section now displays the message:

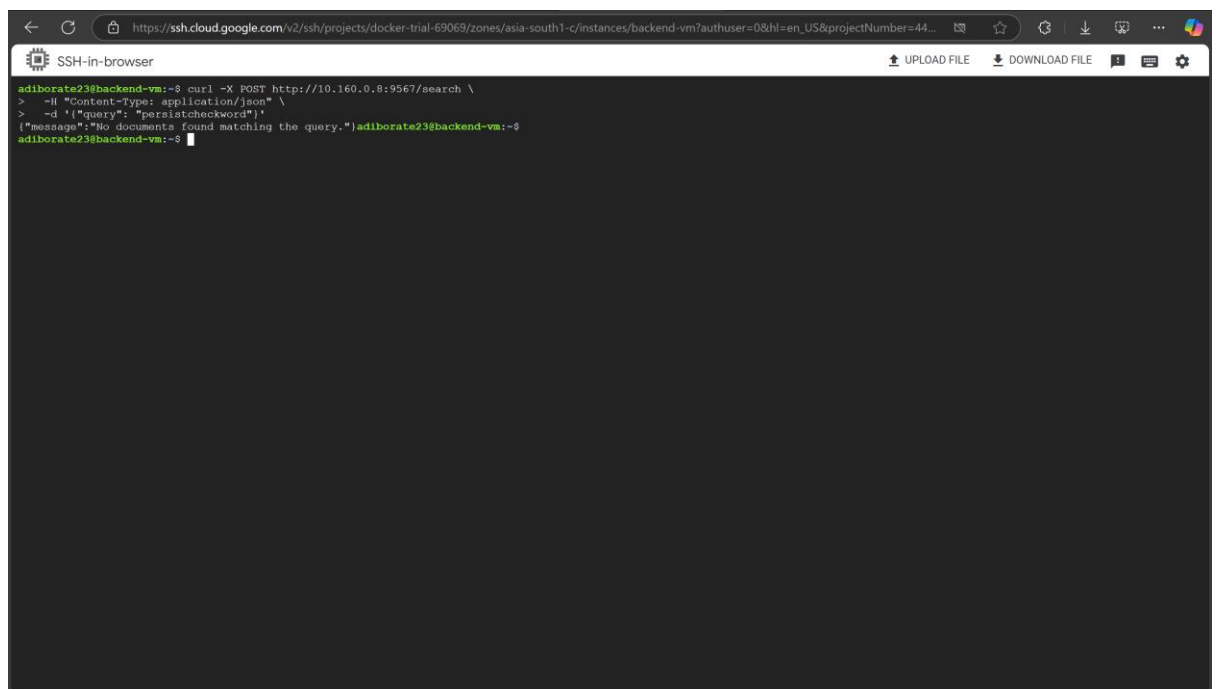
No results to display.



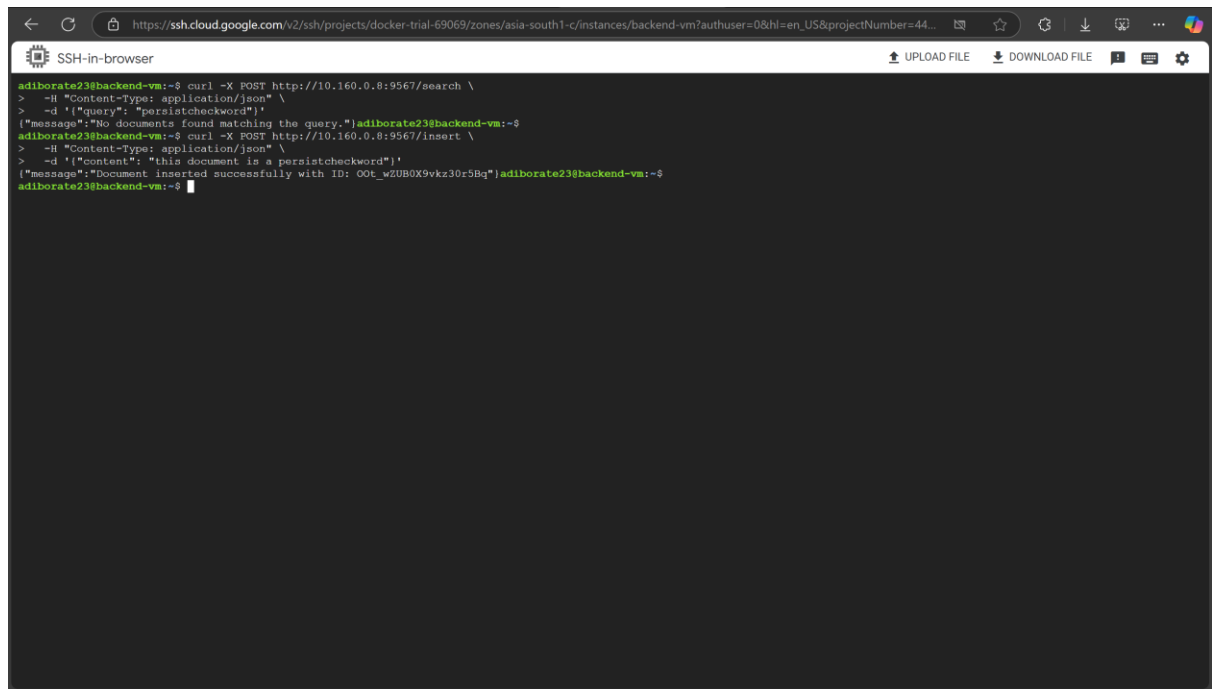
IT WORKED! 👍

✓ Checking data persistence by deleting container

1. Initial Search (Before Insert)



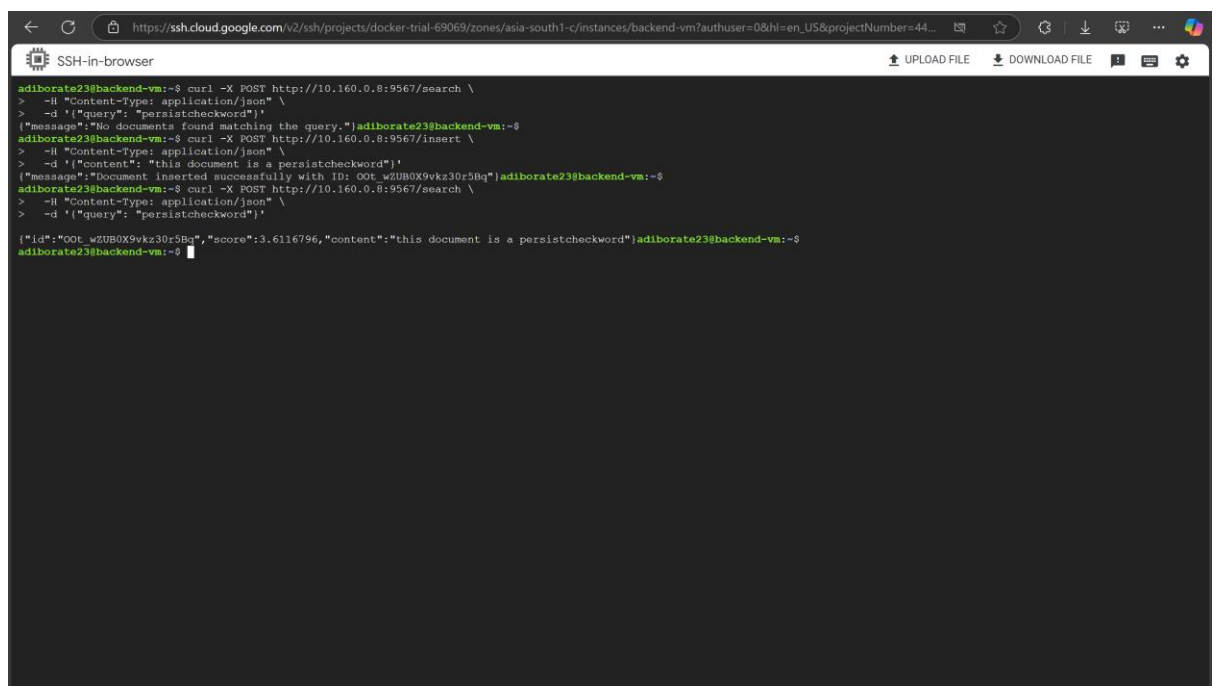
2. Insert a Document



The screenshot shows a terminal window titled "SSH-in-browser" with a URL bar containing a Google Cloud SSH session link. The terminal output shows a series of curl commands and their responses. The first command is a search for "persistcheckword", which returns a message stating "No documents found matching the query.". The second command is an insert of a document with the same content, which returns a message stating "Document inserted successfully with ID: OOT_w2UB0X9vkz30r5Bq".

```
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"message": "No documents found matching the query."}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/insert \
> -H "Content-Type: application/json" \
> -d '{"content": "this document is a persistcheckword"}'
{"message": "Document inserted successfully with ID: OOT_w2UB0X9vkz30r5Bq"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$
```

3. Search Again (After Insert)



The screenshot shows the same terminal window as before, but with an additional search command. The third command is a search for "persistcheckword", which now returns a message containing the document ID, score, and content: {"id": "OOT_w2UB0X9vkz30r5Bq", "score": 3.6116796, "content": "this document is a persistcheckword"}.

```
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"message": "No documents found matching the query."}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/insert \
> -H "Content-Type: application/json" \
> -d '{"content": "this document is a persistcheckword"}'
{"message": "Document inserted successfully with ID: OOT_w2UB0X9vkz30r5Bq"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"id": "OOT_w2UB0X9vkz30r5Bq", "score": 3.6116796, "content": "this document is a persistcheckword"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$
```

4. Stopping and removing all the containers

```
SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...

adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"message": "No documents found matching the query."}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/insert \
> -H "Content-Type: application/json" \
> -d '{"content": "this document is a persistcheckword"}'
{"message": "Document inserted successfully with ID: OOT_wZUB0X9vkz30r5Bq"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"id": "OOT_wZUB0X9vkz30r5Bq", "score": 3.6116796, "content": "this document is a persistcheckword"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
2cb7e3215c82   fastapi:1.2   "python main.py"        9 hours ago    Up 9 hours    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp    lab08-backend-1
elab4390bc71   elasticsearch  "/bin/tini -- /usr/l..." 9 hours ago    Up 9 hours    9200/tcp, 9300/tcp                    lab08-elasticsearch-1
adiborate23@backend-vm:~$ cd lab08
adiborate23@backend-vm:~/lab08$ docker-compose down
[+] Running 4/1
  ✓ Container lab08-backend-1   Removed
  ✓ Container lab08-setup-1     Removed
  ✓ Container lab08-elasticsearch-1 Removed
  ✓ Network lab08_app-network   Removed
adiborate23@backend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
adiborate23@backend-vm:~/lab08$
```

```
SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/frontend-vm?authuser=0&hl=en_US&projectNumber=44...

adiborate23@frontend-vm:~$ cd lab08
adiborate23@frontend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
3d67858f31d7   fastapi:1.1   "python main.py"        24 hours ago    Up 24 hours    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp    lab08-frontend-1
adiborate23@frontend-vm:~/lab08$ docker-compose down
[+] Running 2/2
  ✓ Container lab08-frontend-1   Removed
  ✓ Network lab08_default        Removed
adiborate23@frontend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
adiborate23@frontend-vm:~/lab08$
```

5. Restarting Containers

```

SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...

adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"message": "No documents found matching the query."}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/insert \
> -H "Content-Type: application/json" \
> -d '{"content": "this document is a persistcheckword"}'
{"message": "Document inserted successfully with ID: OOT_wZUBOX9vkz30r5Bq"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"id": "OOT_wZUBOX9vkz30r5Bq", "score": 3.6116796, "content": "this document is a persistcheckword"}adiborate23@backend-vm:~$
adiborate23@backend-vm:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
2cb7e3215c82   fastapi:1.2   "python main.py"        9 hours ago    Up 9 hours    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp    lab08-backend-1
elab4390bc71   elasticsearch   "/bin/tini -- /usr/l..." 9 hours ago    Up 9 hours    9200/tcp, 9300/tcp                    lab08-elasticsearch-1
adiborate23@backend-vm:~$ ls lab08
backend  docker-compose.yml  elasticsearch  setup
adiborate23@backend-vm:~$ cd lab08
adiborate23@backend-vm:~/lab08$ docker-compose down
[+] Running 4/4
  ✓ Container lab08-backend-1   Removed
  ✓ Container lab08-setup-1     Removed
  ✓ Container lab08-elasticsearch-1 Removed
  ✓ Network lab08_app-network   Removed
adiborate23@backend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
adiborate23@backend-vm:~/lab08$ docker-compose up -d
[+] Running 4/4
  ✓ Network lab08_app-network   Created
  ✓ Container lab08-elasticsearch-1 Healthy
  ✓ Container lab08-setup-1     Started
  ✓ Container lab08-backend-1   Started
adiborate23@backend-vm:~/lab08$

```

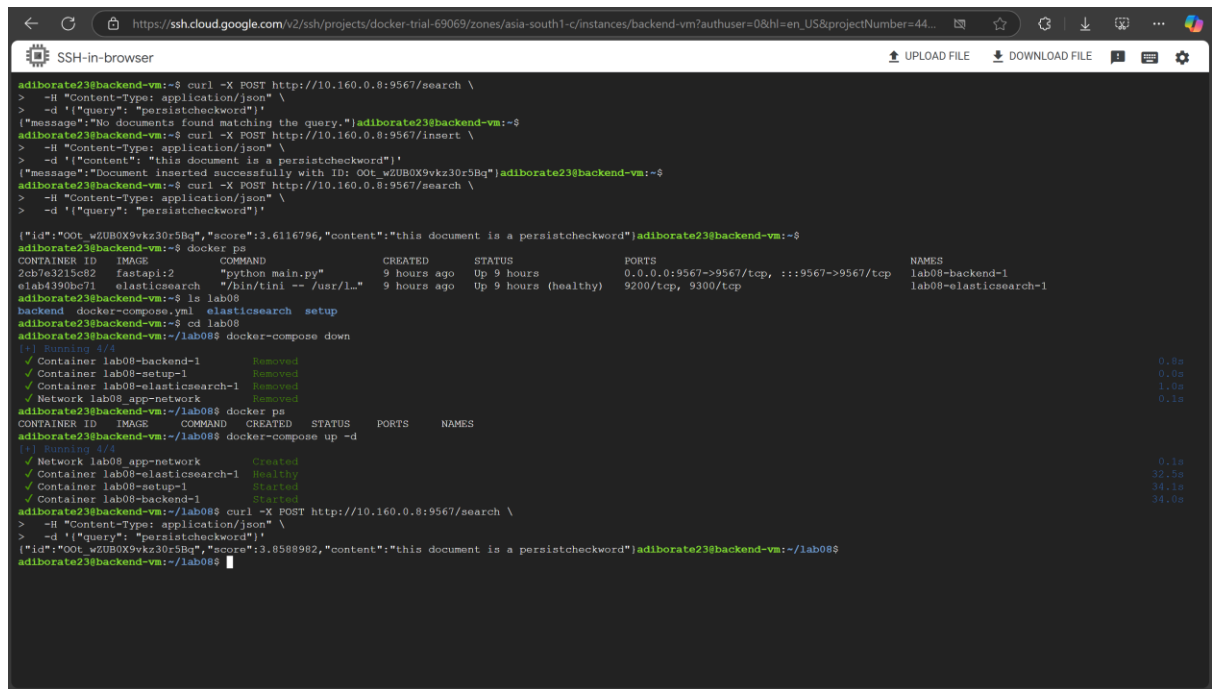
```

SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/frontend-vm?authuser=0&hl=en_US&projectNumber=44...

adiborate23@frontend-vm:~$ cd lab08
adiborate23@frontend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
3d67858f31d7   fastapi:1.2   "python main.py"        24 hours ago    Up 24 hours    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp    lab08-frontend-1
adiborate23@frontend-vm:~/lab08$ docker-compose down
[+] Running 2/2
  ✓ Container lab08-frontend-1   Removed
  ✓ Network lab08 default        Removed
adiborate23@frontend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
adiborate23@frontend-vm:~/lab08$ docker-compose up -d
[+] Running 2/2
  ✓ Network lab08 default        Created
  ✓ Container lab08-frontend-1   Started
adiborate23@frontend-vm:~/lab08$

```

6. Final Search (After Restarting)



```

adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"message": "No documents found matching the query."}
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/insert \
> -H "Content-Type: application/json" \
> -d '{"content": "this document is a persistcheckword"}'
{"message": "Document inserted successfully with ID: OOT_wZUB0X9vkz30r5Bq"}
adiborate23@backend-vm:~$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"id": "OOT_wZUB0X9vkz30r5Bq", "score": 3.6116796, "content": "this document is a persistcheckword"}
adiborate23@backend-vm:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
2cb7e3215c82   fastapi:2     "python main.py"        9 hours ago    Up 9 hours    0.0.0.0:9567->9567/tcp, :::9567->9567/tcp    lab08-backend-1
adiborate23@backend-vm:~$ ls lab08
backend  docker-compose.yml  elasticsearch  setup
adiborate23@backend-vm:~$ cd lab08
adiborate23@backend-vm:~/lab08$ docker-compose down
[+] Running 4/4
  ✓ Container lab08-backend-1   Removed
  ✓ Container lab08-setup-1     Removed
  ✓ Container lab08-elasticsearch-1 Removed
  ✓ Network lab08_app-network   Removed
adiborate23@backend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
adiborate23@backend-vm:~/lab08$ docker-compose up -d
[+] Running 4/4
  ✓ Network lab08_app-network   Created
  ✓ Container lab08-elasticsearch-1 Healthy
  ✓ Container lab08-setup-1     Started
  ✓ Container lab08-backend-1   Started
adiborate23@backend-vm:~/lab08$ curl -X POST http://10.160.0.8:9567/search \
> -H "Content-Type: application/json" \
> -d '{"query": "persistcheckword"}'
{"id": "OOT_wZUB0X9vkz30r5Bq", "score": 3.8588982, "content": "this document is a persistcheckword"}
adiborate23@backend-vm:~/lab08$

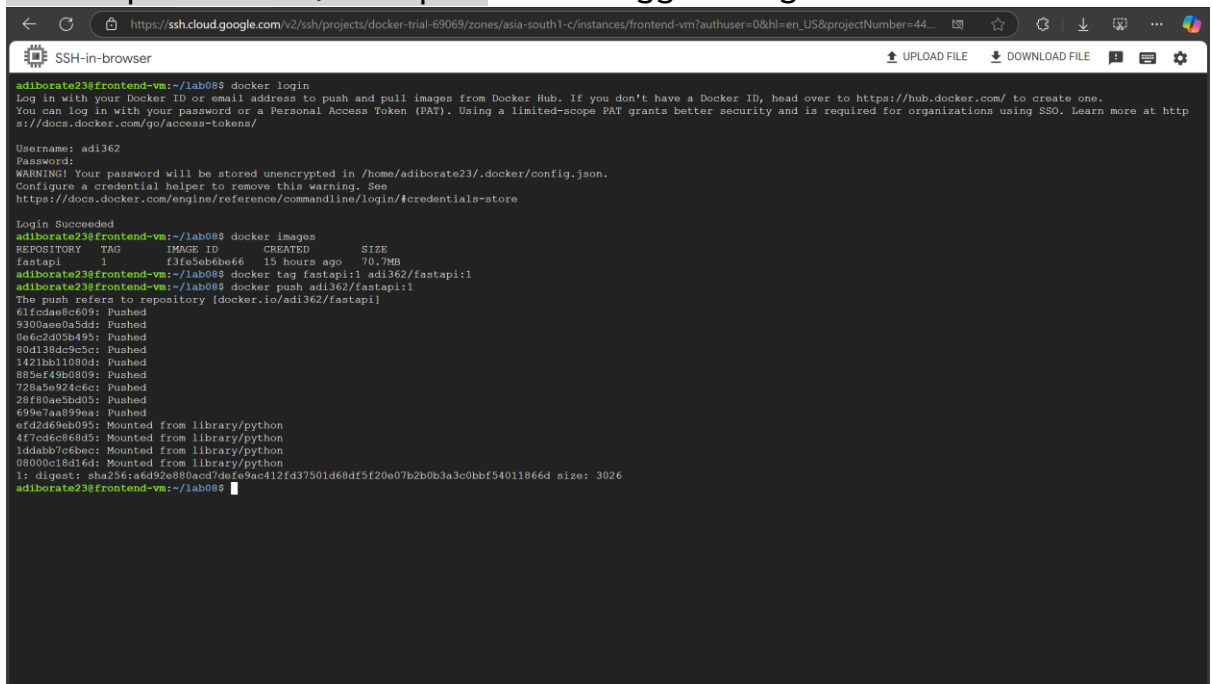
```

This confirms that data has persisted through container deletion.

✓ Uploading images created for all three tasks

- From frontend-vm:

1. `docker login`: to log in into dockerhub account
2. `docker tag fastapi:1 adi362/fastapi:1` : Tag images with my username
3. `docker push adi362/fastapi:1`: Push tagged images to Docker Hub



```

adiborate23@frontend-vm:~/lab08$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/
Username: adi362
Password:
WARNING! Your password will be stored unencrypted in /home/adiborate23/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
adiborate23@frontend-vm:~/lab08$ docker images
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
fastapi      1         f3fc8eb0a66   15 hours ago   70.7MB
adiborate23@frontend-vm:~/lab08$ docker tag fastapi:1 adi362/fastapi:1
adiborate23@frontend-vm:~/lab08$ docker push adi362/fastapi:1
The push refers to repository [docker.io/adi362/fastapi]
61fcdac8c09: Pushed
9300ae0a5d1: Pushed
9e6c2d02b49: Pushed
80d138dc9c5: Pushed
1421bb1109d: Pushed
895ef49b080: Pushed
728a5e924dc: Pushed
28f80ae5bd0: Pushed
699e7aa899ea: Pushed
efd2d69eb09: Mounted from library/python
4f7ed6cfe605: Mounted from library/python
1ddabb7c6bae: Mounted from library/python
08000c18d16d: Mounted from library/python
1: digest: sha256:a6d92e880acd7dfe9ac412fd37501d68df520e07b2b0b3ac0bbf54011866d size: 3026
adiborate23@frontend-vm:~/lab08$

```

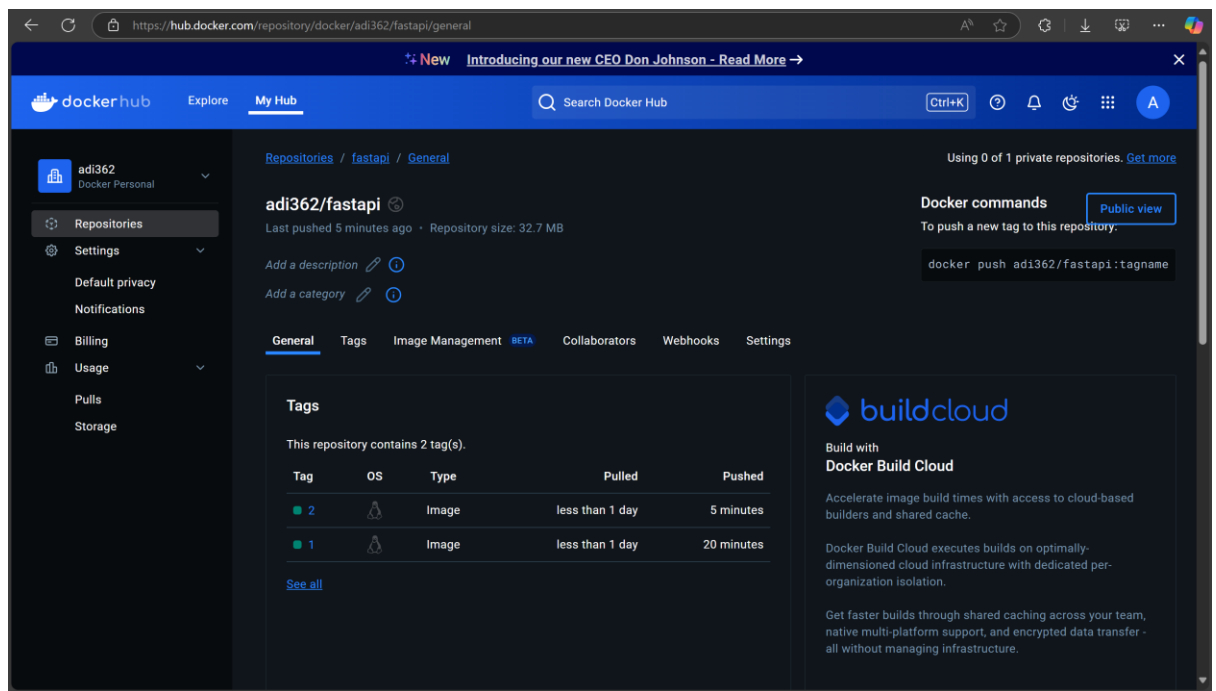
- Similarly from backend-vm:


```
SSH-in-browser
https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...
UPLOAD FILE DOWNLOAD FILE

adiborate23@backend-vm:~/lab08$ docker images
REPOSITORY    TAG       IMAGE ID      CREATED      SIZE
fastapi        2         f5b089909907  3 hours ago  73.3MB
setup          latest    2c14e931ae3c  14 hours ago  61.1MB
elasticsearch  latest    c950f92fcbab  14 hours ago  1.04GB
adiborate23@backend-vm:~/lab08$ docker tag fastapi:2 adi362/fastapi:2
adiborate23@backend-vm:~/lab08$ docker tag elasticsearch:latest adi362/elasticsearch:latest
adiborate23@backend-vm:~/lab08$ docker tag setup:latest adi362/setup:latest
adiborate23@backend-vm:~/lab08$ docker push adi362/fastapi:2
The push refers to repository [docker.io/adi362/fastapi]
af863b28bcb1b: Pushed
f5bc1b299ffa: Pushed
1a2398a503cd: Pushed
d512f327e3ab: Pushed
d85cbb96d8fb: Pushed
0a803c94067d: Pushed
ab7125fcb19c: Pushed
891b77854b75: Pushed
afd2d69ab095: Layer already exists
4f7cd6c8e6d5: Layer already exists
1ddabb7c6bec: Layer already exists
08000c18d16d: Layer already exists
2: digest: sha256:101c70aa921ee17012e047d164909ae2d848474c38871c680987830fa4e0e813 size: 2819
adiborate23@backend-vm:~/lab08$ docker push adi362/elasticsearch:latest
The push refers to repository [docker.io/adi362/elasticsearch]
c97a3d634b80: Pushed
0d1a9d397b68: Pushed
2ef148f1d4c4: Mounted from library/elasticsearch
d4e7d11fb04d: Mounted from library/elasticsearch
d8fb75b0ae1f: Mounted from library/elasticsearch
e1f0773e8580: Mounted from library/elasticsearch
77db73b69c5c: Mounted from library/elasticsearch
bc73a95c0b43: Mounted from library/elasticsearch
2653d992f4ef: Mounted from library/elasticsearch
latest: digest: sha256:b4156b065aa6252a659e05190b3e0e18019392e05ece152c5c491327763b9d4e size: 2202
adiborate23@backend-vm:~/lab08$ docker push adi362/setup:latest
The push refers to repository [docker.io/adi362/setup]
9f9bc34f9898: Pushed
bc739b21ae63: Pushed
0afdfabb8d1: Pushed
891b77854b75: Mounted from ad362/fastapi
afd2d69ab095: Mounted from ad362/fastapi
4f7cd6c8e6d5: Mounted from ad362/fastapi
1ddabb7c6bec: Mounted from ad362/fastapi
08000c18d16d: Mounted from ad362/fastapi
latest: digest: sha256:cfc6d47e2cd15ffe0afc8e264a6329ab406aebd8e0db613bff436fb761ded317 size: 1989
adiborate23@backend-vm:~/lab08$
```

4. DockerHub Repositories

The screenshot displays the Docker Hub 'My Hub' page for the user 'adi362'. The interface includes a sidebar with navigation links: 'Repositories', 'Settings', 'Default privacy', 'Notifications', 'Billing', 'Usage', 'Pulls', and 'Storage'. The main content area, titled 'Repositories', shows a list of repositories within the 'adi362' namespace. The list includes three repositories: 'adi362/setup' (pushed 2 minutes ago), 'adi362/elasticsearch' (pushed 4 minutes ago), and 'adi362/fastapi' (pushed 5 minutes ago). Each repository entry features a status icon (labeled 'IMAGE'), a 'Public' visibility status, and an 'Inactive' Scout status. A search bar and a 'Create a repository' button are also visible at the top of the repository list.



✓ Optimization Checks

- Using python:3.9-alpine — that's already a minimal base image.
- Pip installs are done with --no-cache-dir — avoids unnecessary bloat.
- Non-root user — great security practice.
- Only necessary files are copied — no extra layers or unnecessary bloating.
- No extra tools/libraries or debugging stuff in the final image.
- SSL certs handled properly and permissions adjusted.

✓ Detailed Architecture

